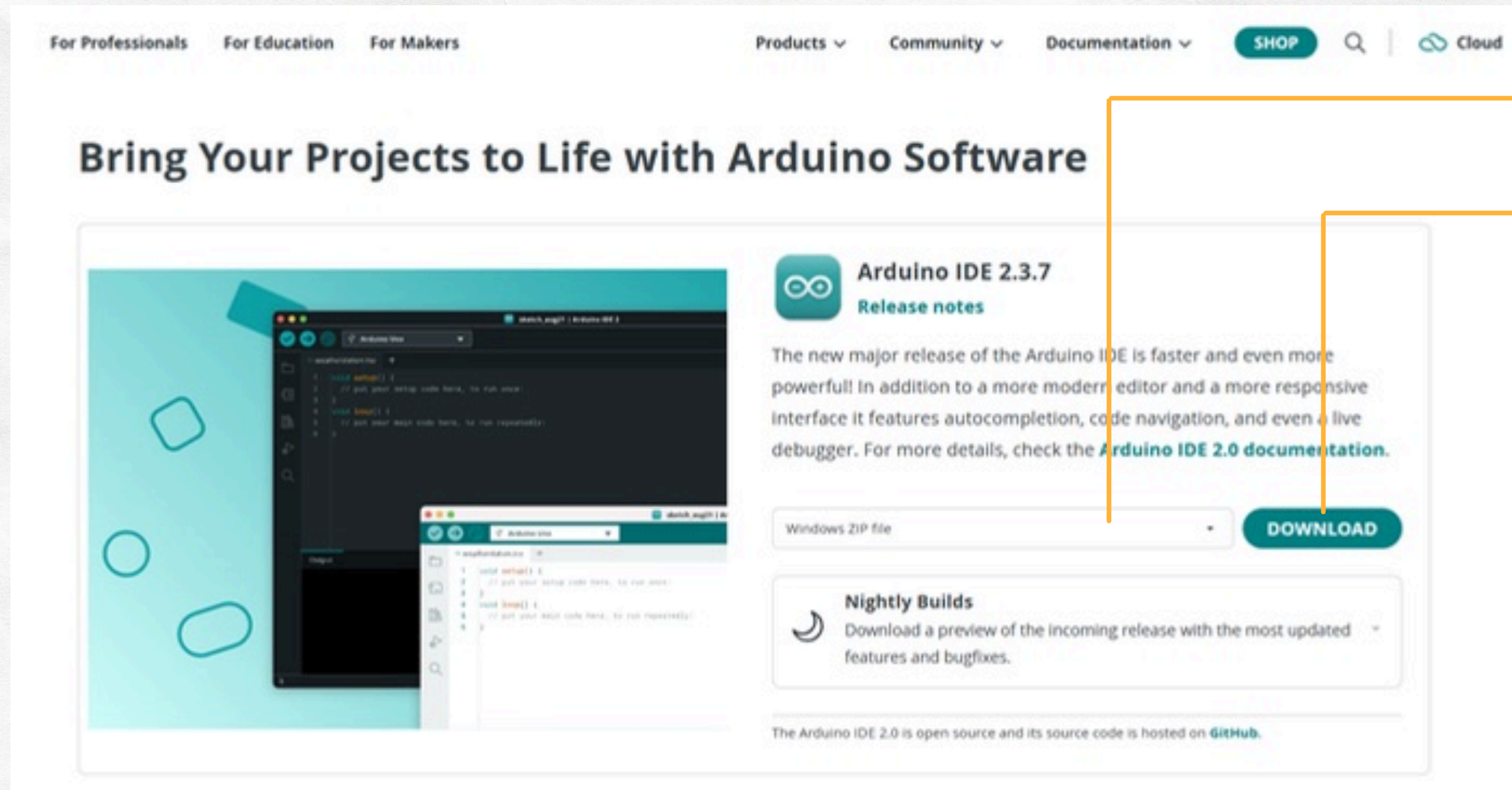


Welcome to Mechanical Flower Workshop!

Please install the **Arduino IDE** first.

<https://www.arduino.cc/en/software/> **Project Github Link:** <https://shorturl.at/78yy2>



① Choose the version that suits your computer.

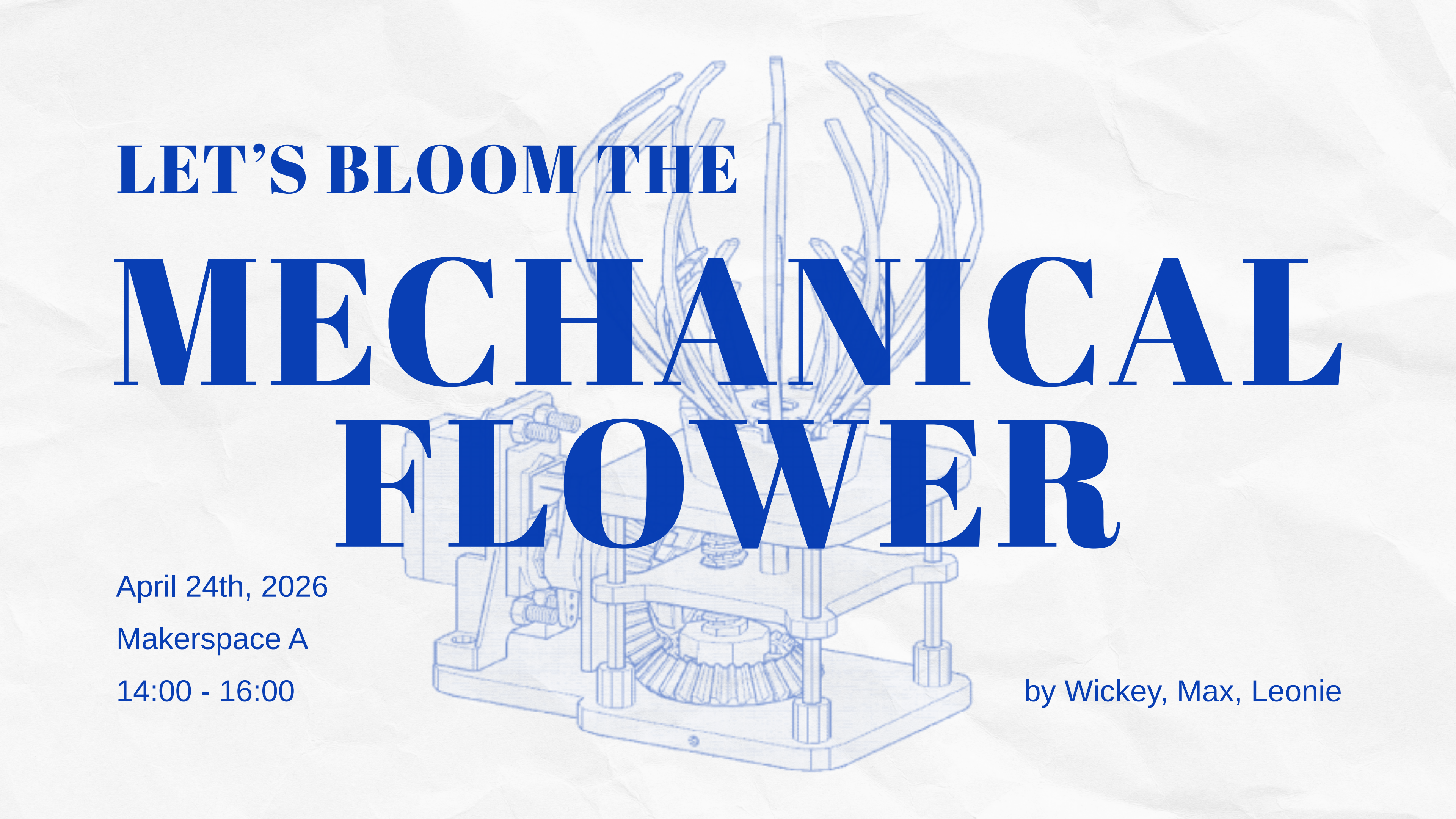
② Download and install.

Arduino IDE



MacOS needs to install CH340 Driver

 CH341SER_MAC.ZIP

The background features a detailed blue line drawing of a mechanical flower. The base is a complex assembly of gears, including a large central gear and several smaller ones. Vertical rods support a platform from which multiple thin, wire-like structures rise, resembling flower petals or stamens. The entire illustration is rendered in a light blue, sketch-like style against a textured, light-colored background.

LET'S BLOOM THE MECHANICAL FLOWER

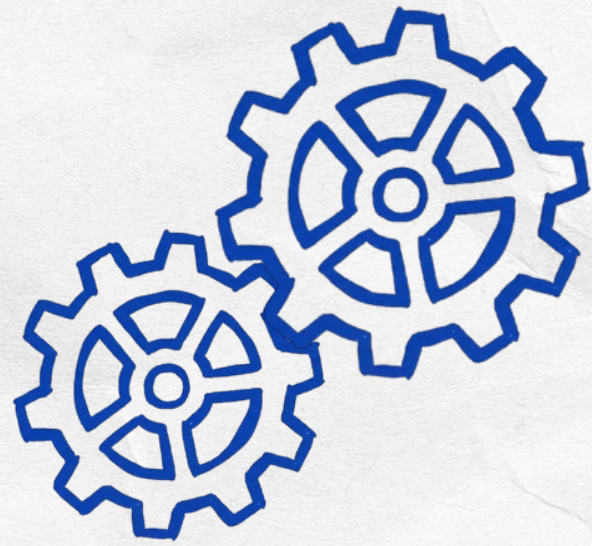
April 24th, 2026

Makerspace A

14:00 - 16:00

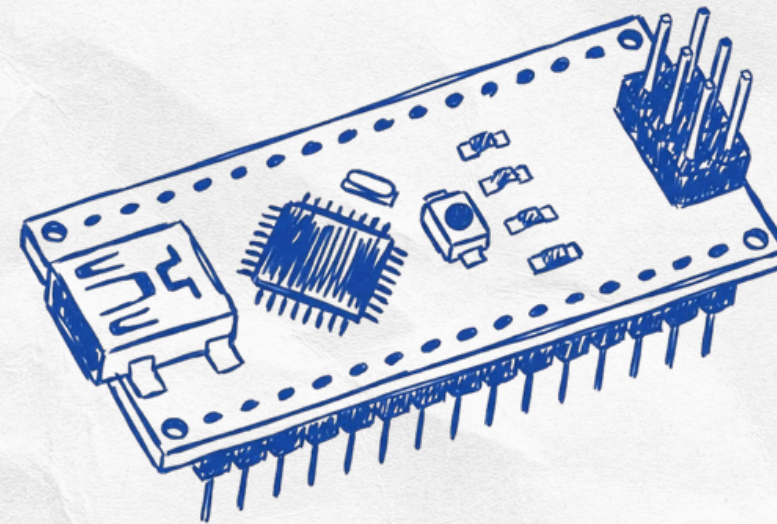
by Wickey, Max, Leonie

introduction



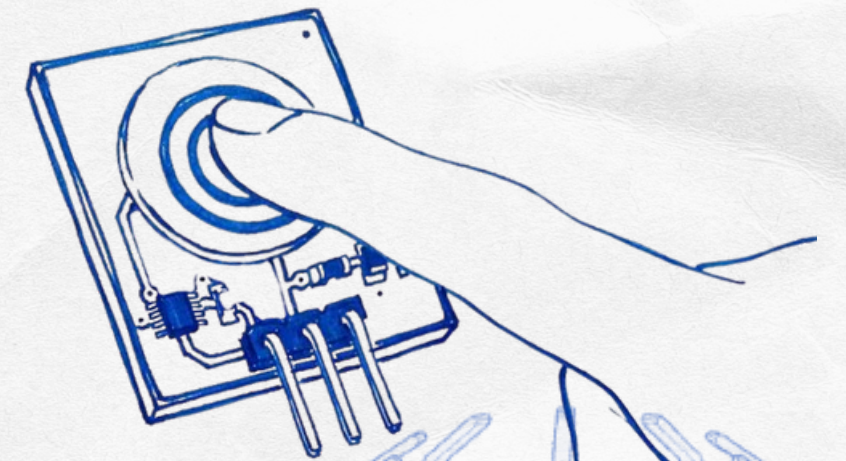
TransMission

How gears drive the flower bloom?
Step-by-step to assemble it!



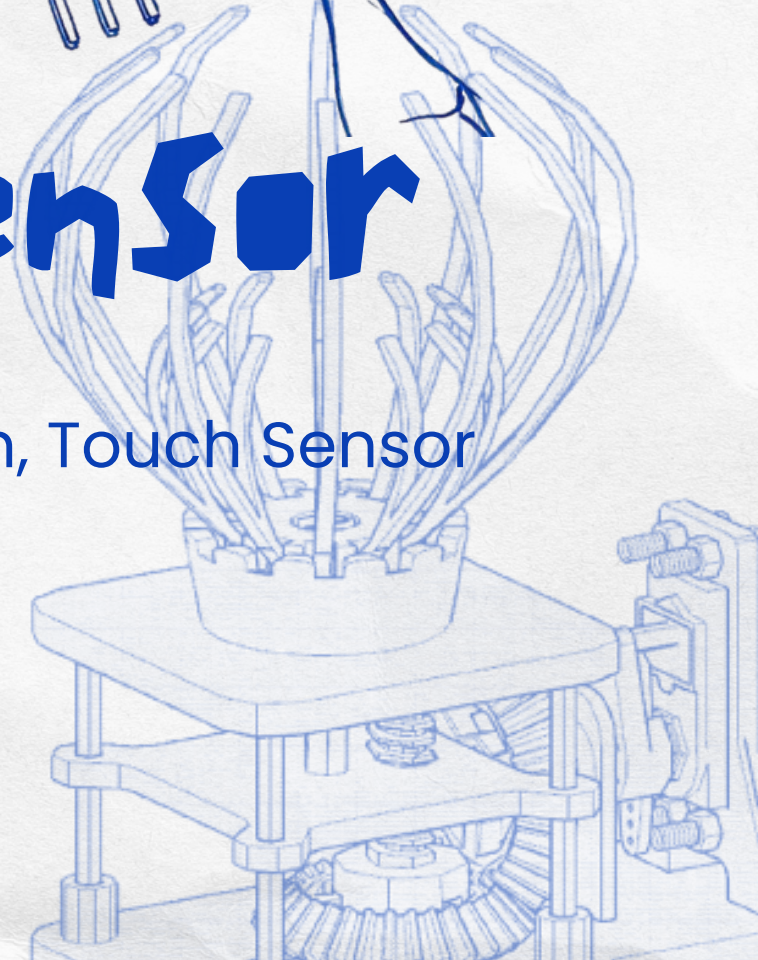
Arduino

Configure the Arduino environment;
Drive the MG996R servo by command.



Sensor

Button, Touch Sensor

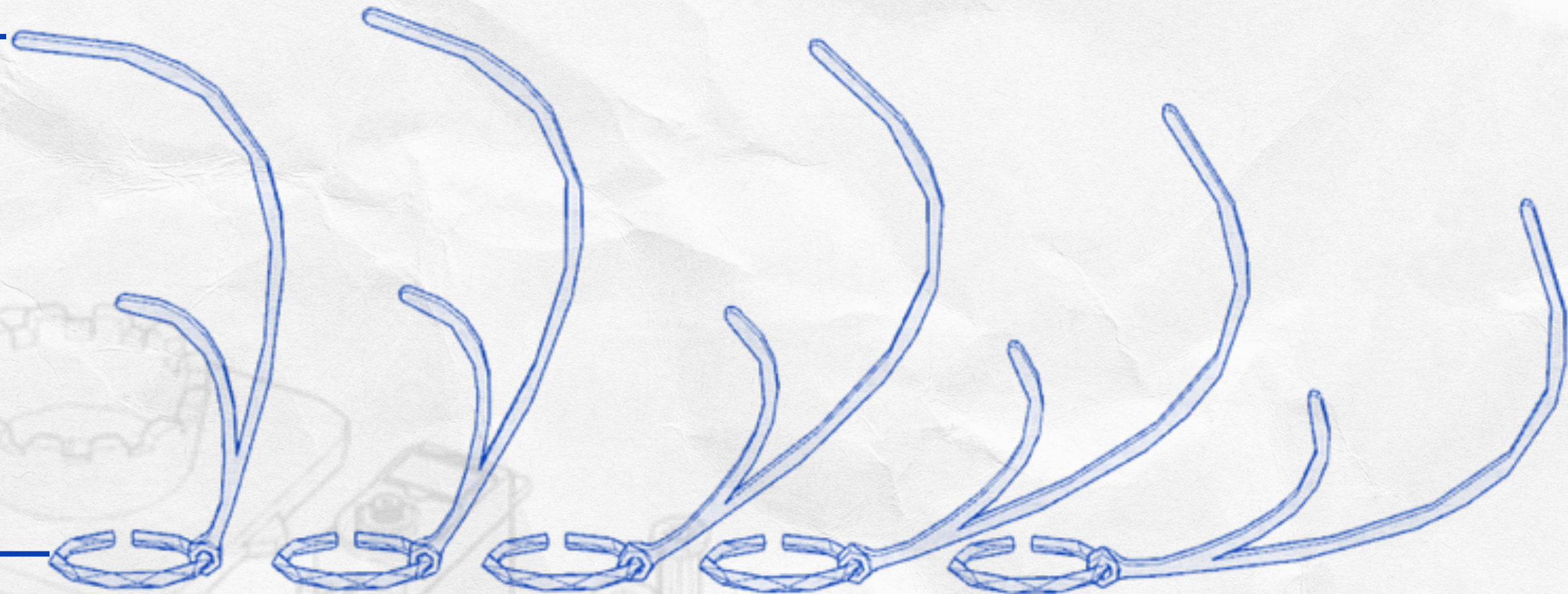


How do the mechanical parts move like flower?

1. Let the petal parts move like a real flower

Petals

Central Axis

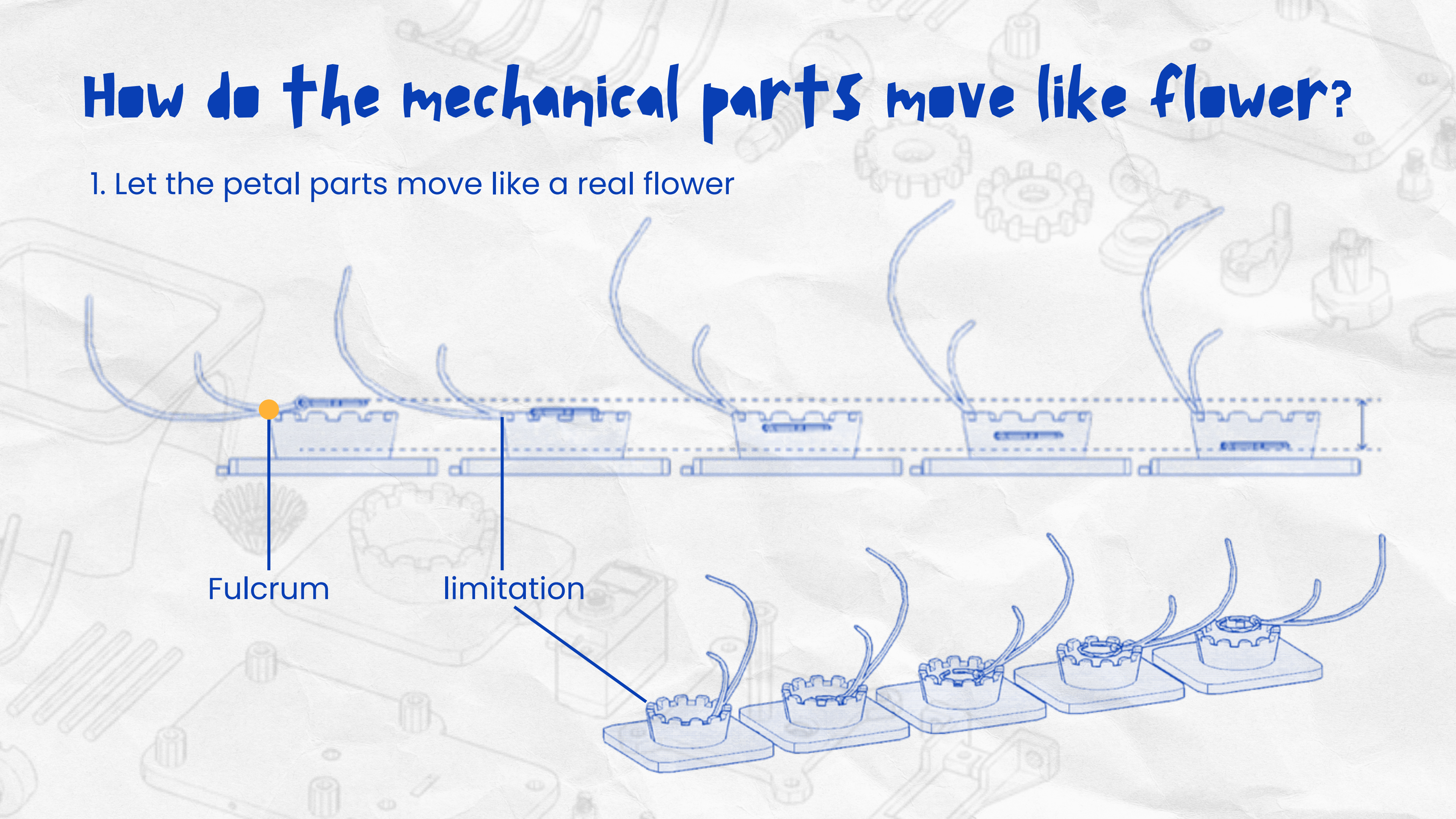


How do the mechanical parts move like flower?

1. Let the petal parts move like a real flower

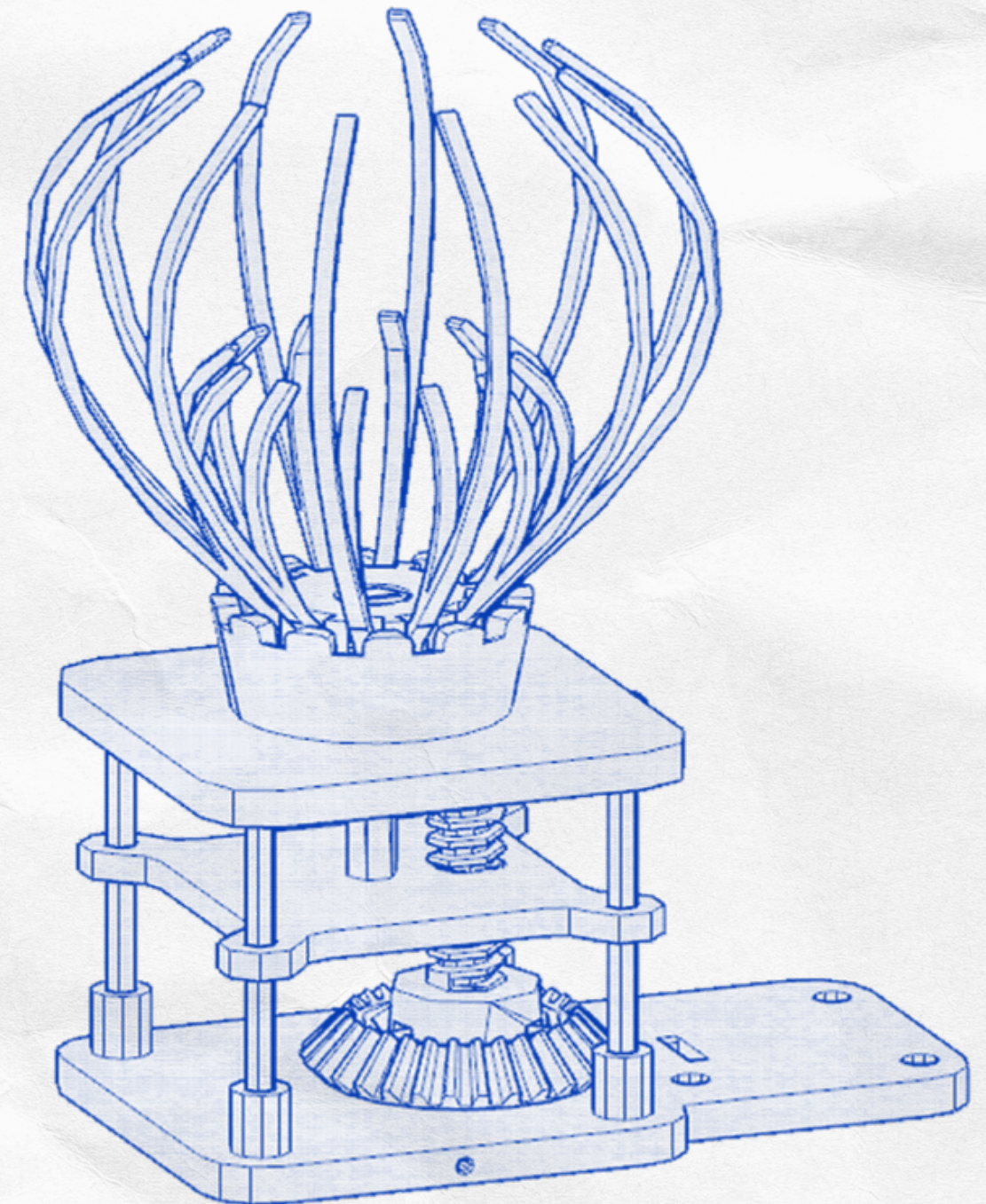
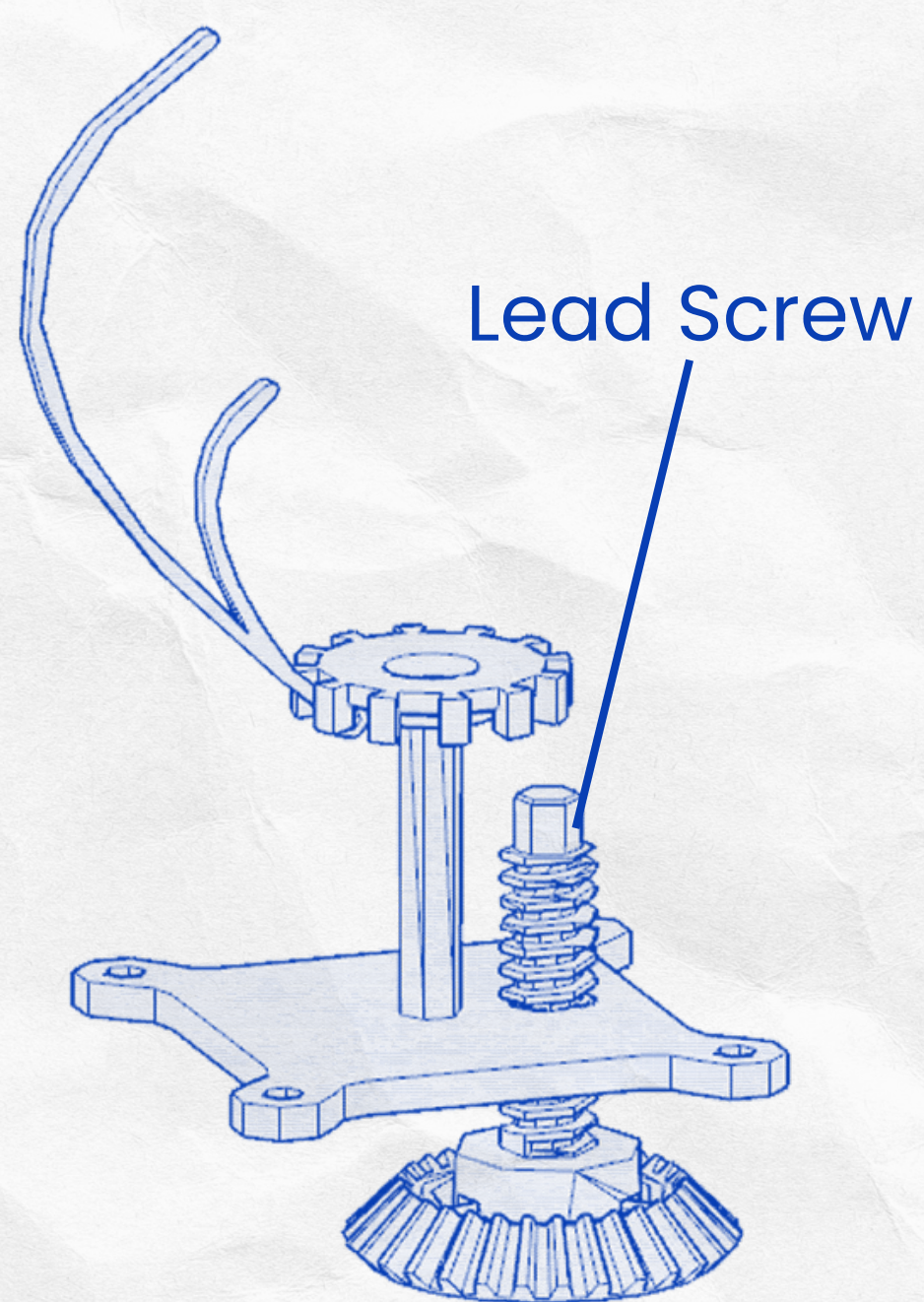
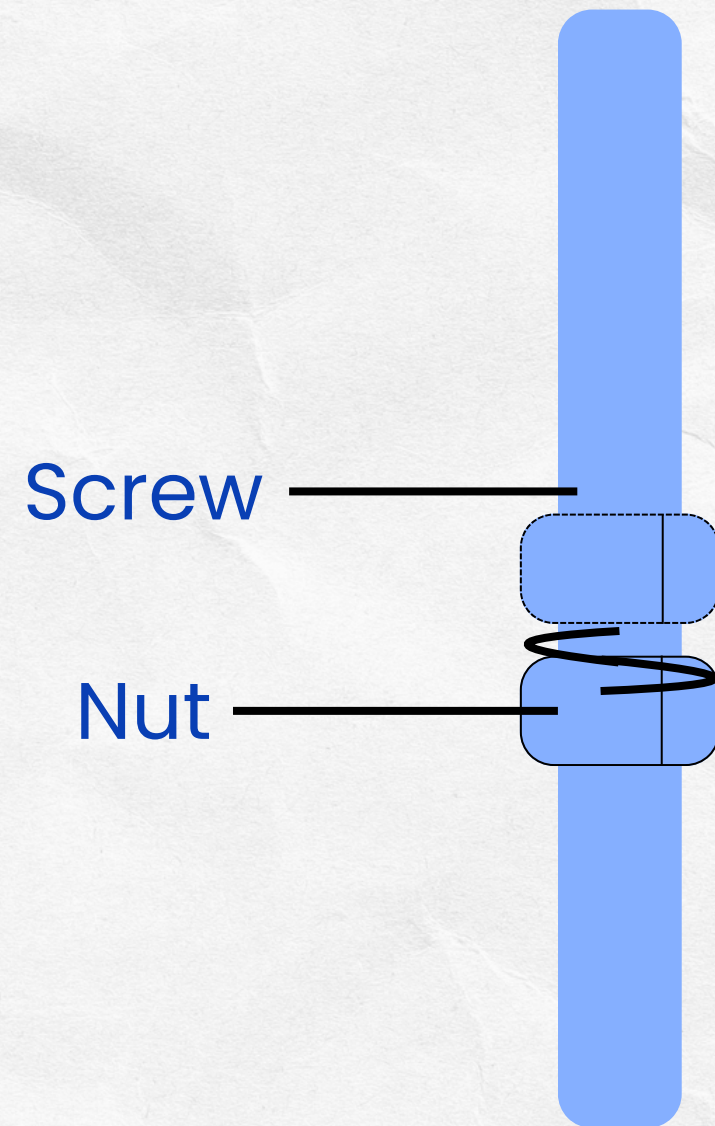
Fulcrum

limitation



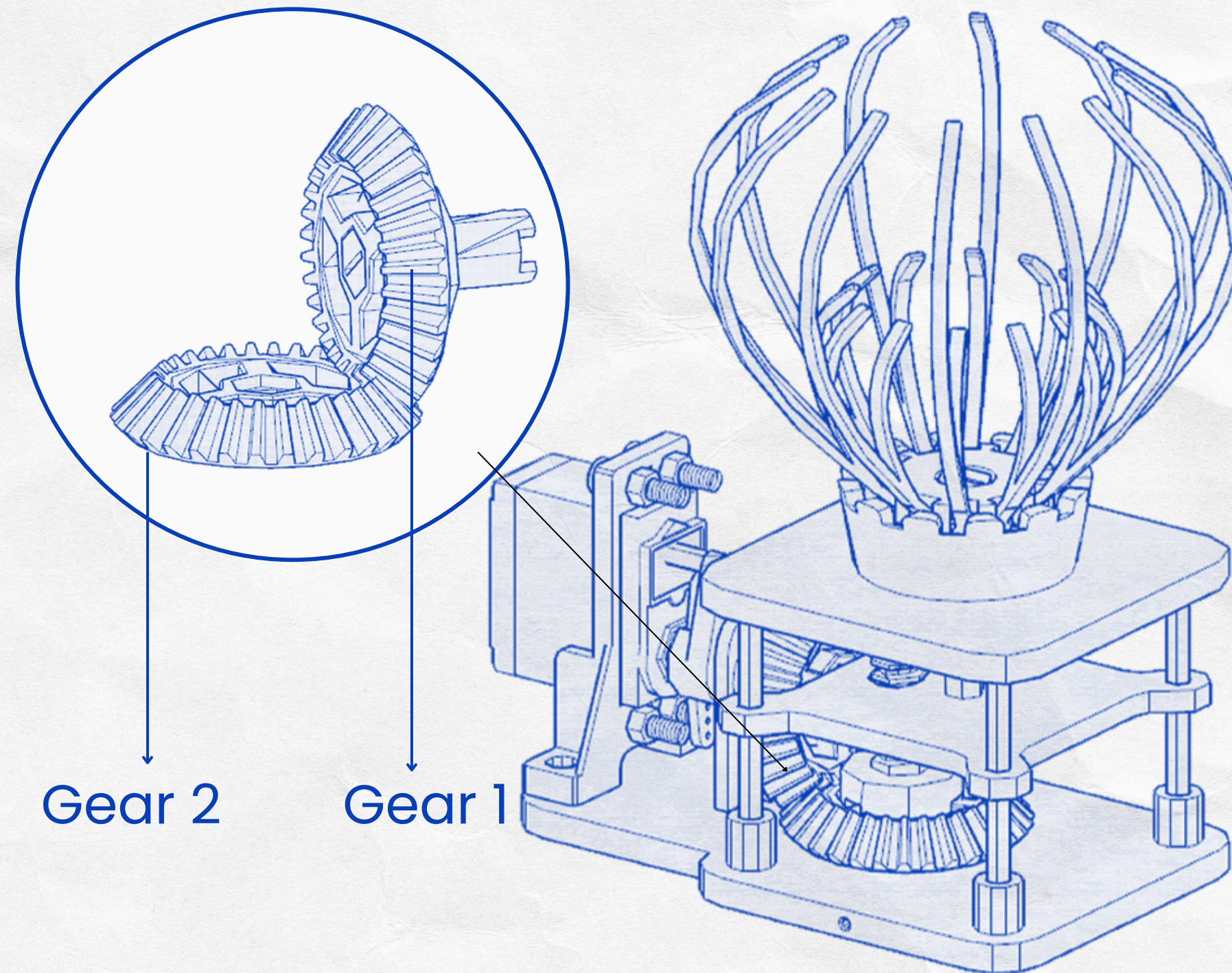
How do the mechanical parts move like flower?

2. How to generate a vertical up-and-down force?



How do the mechanical parts move like flower?

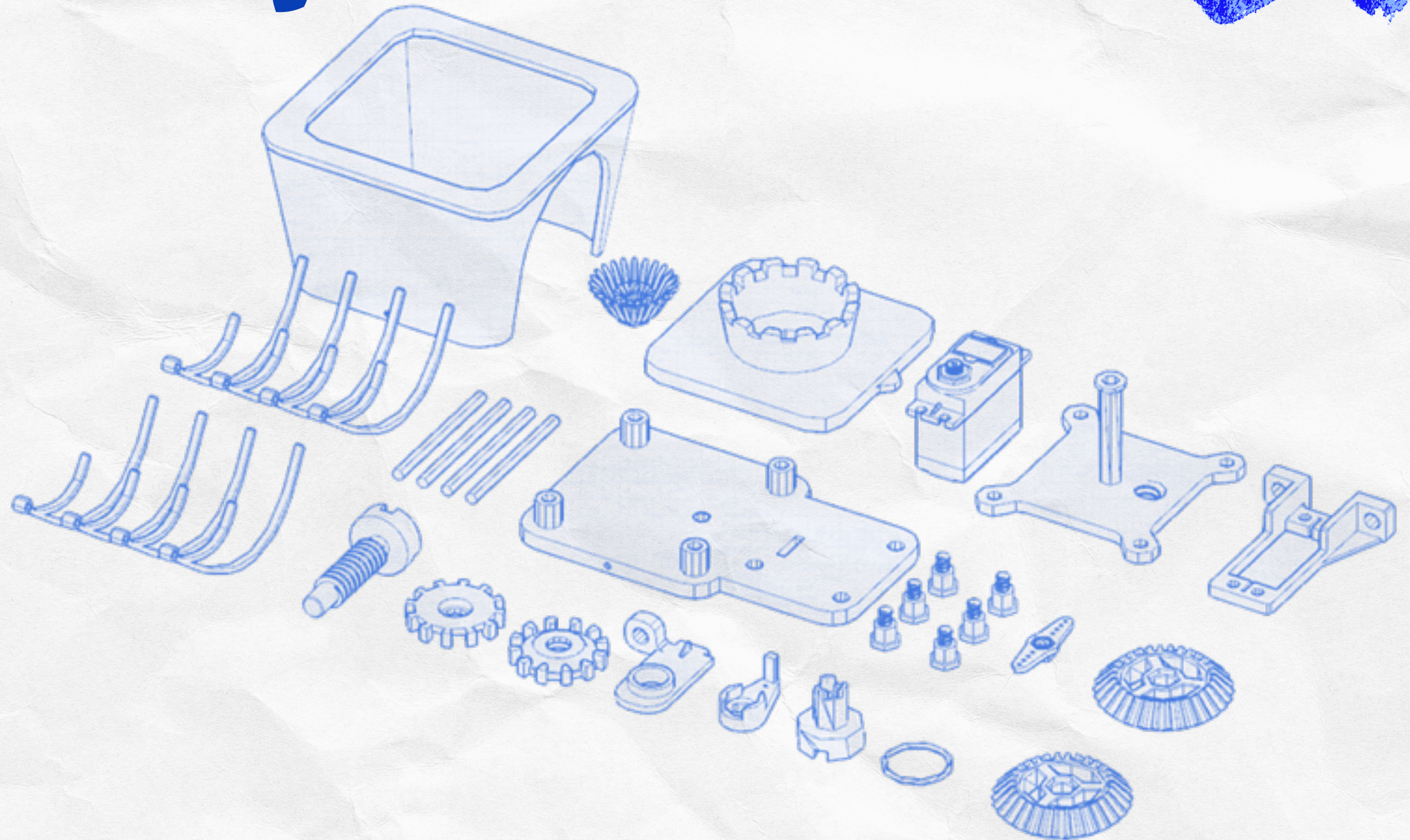
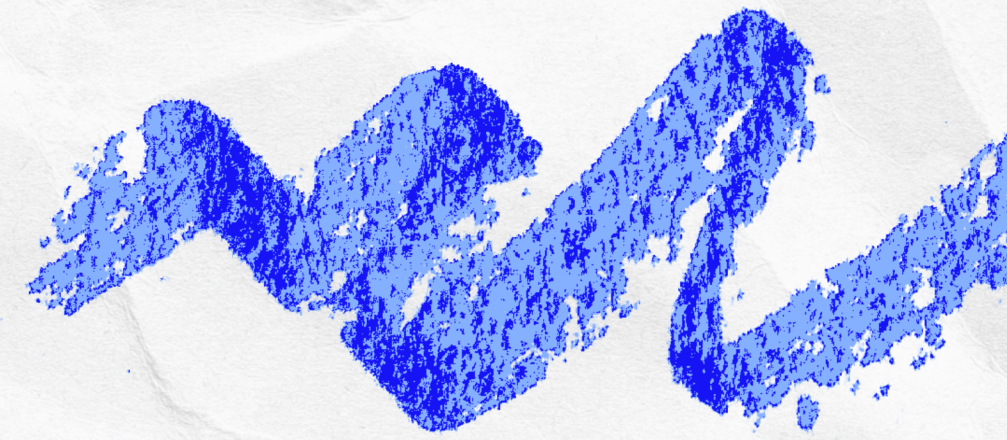
3. Gear transmission



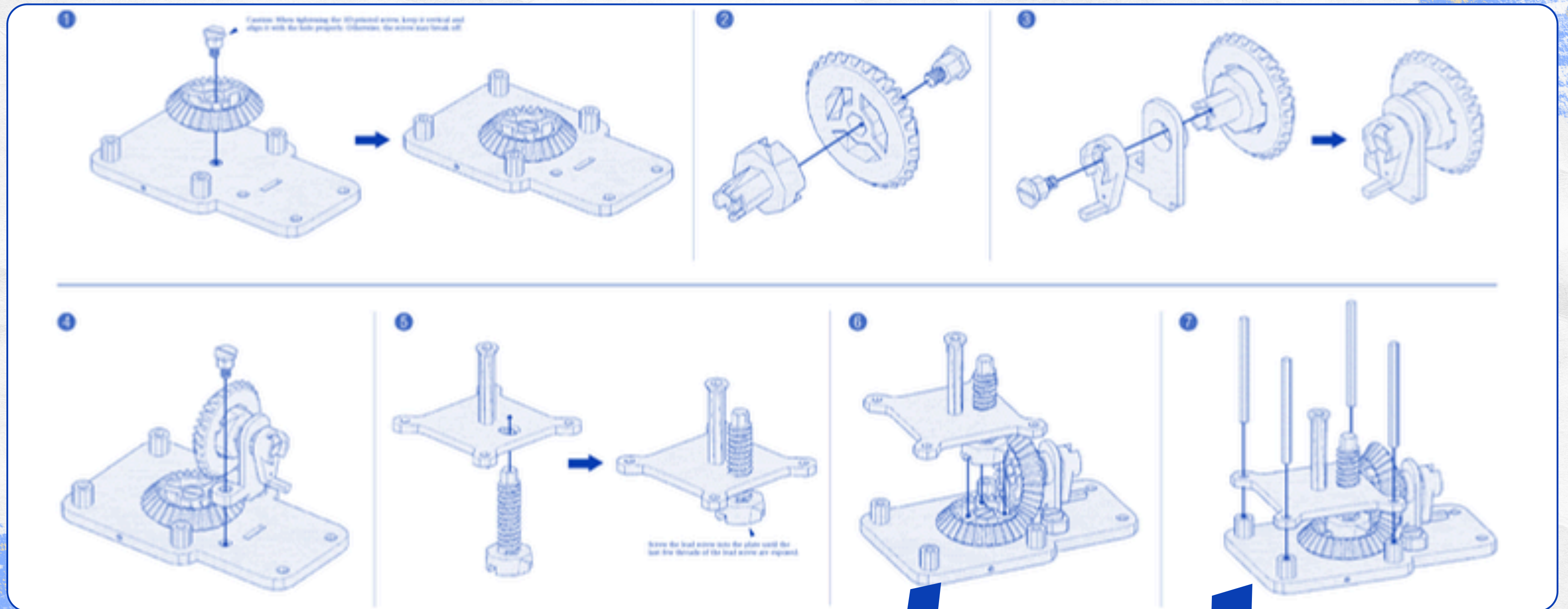
Why?

Change the Direction of Power:
from vertical to horizontal.

3D Printing Parts



Please check the guidance sheet on your table.

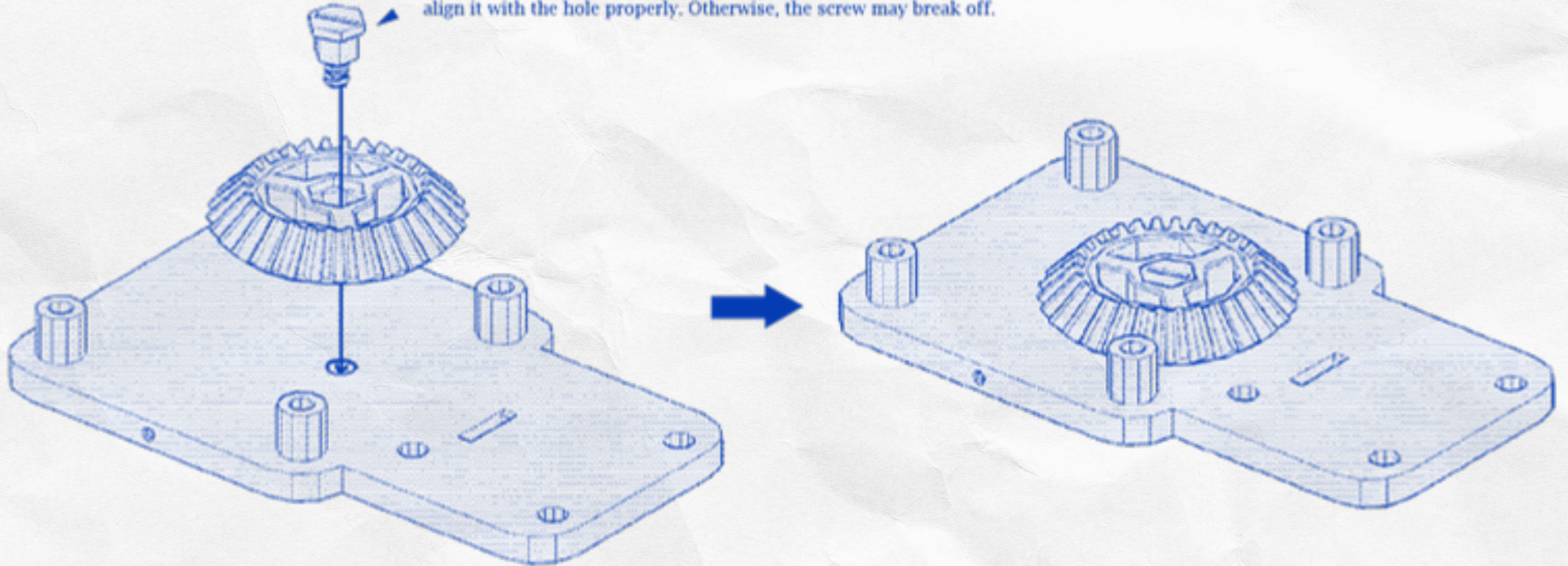


ASSEMBLE it!

Let's ASSEMBle it!

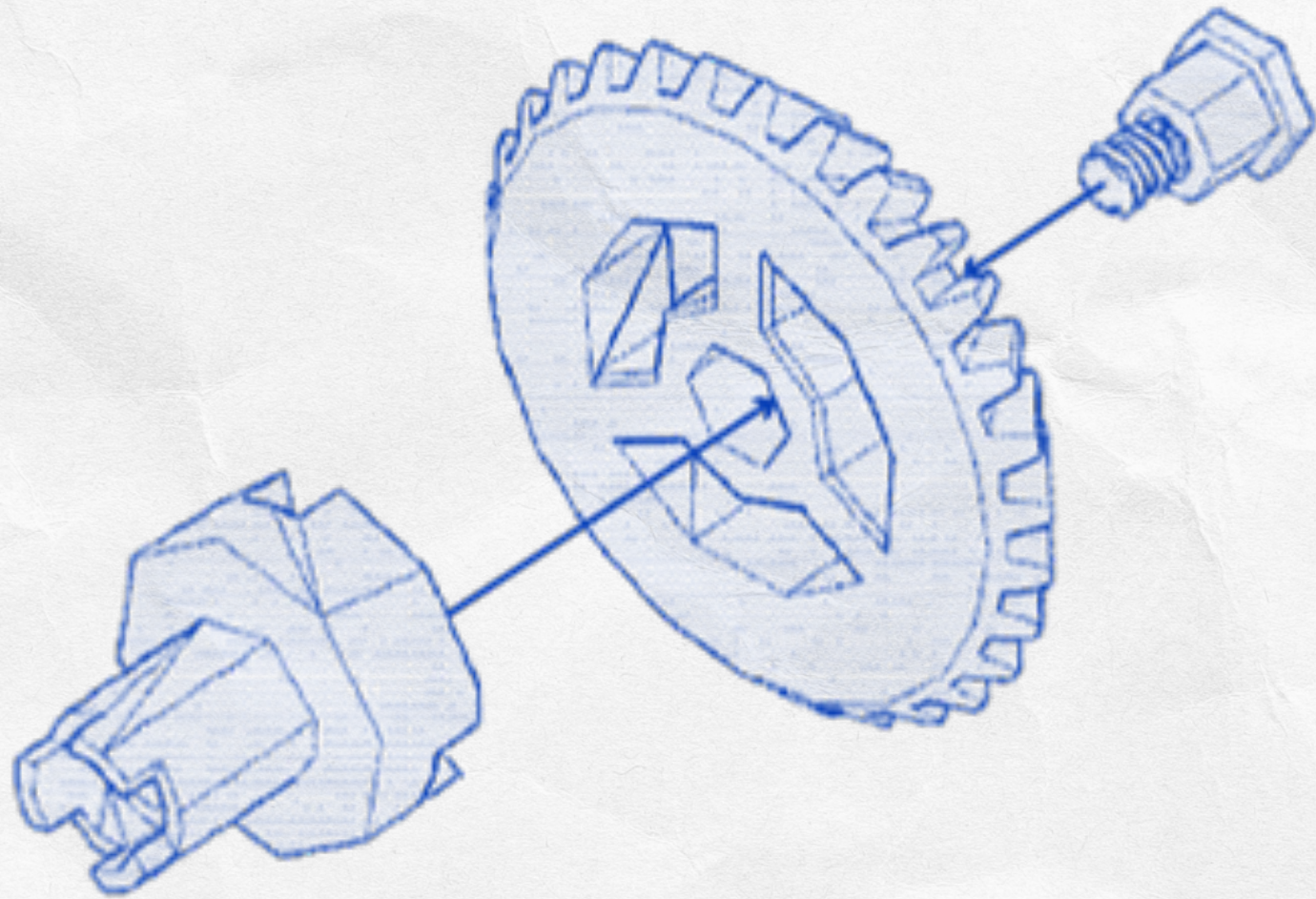
1

Caution: When tightening the 3D-printed screw, keep it vertical and align it with the hole properly. Otherwise, the screw may break off.

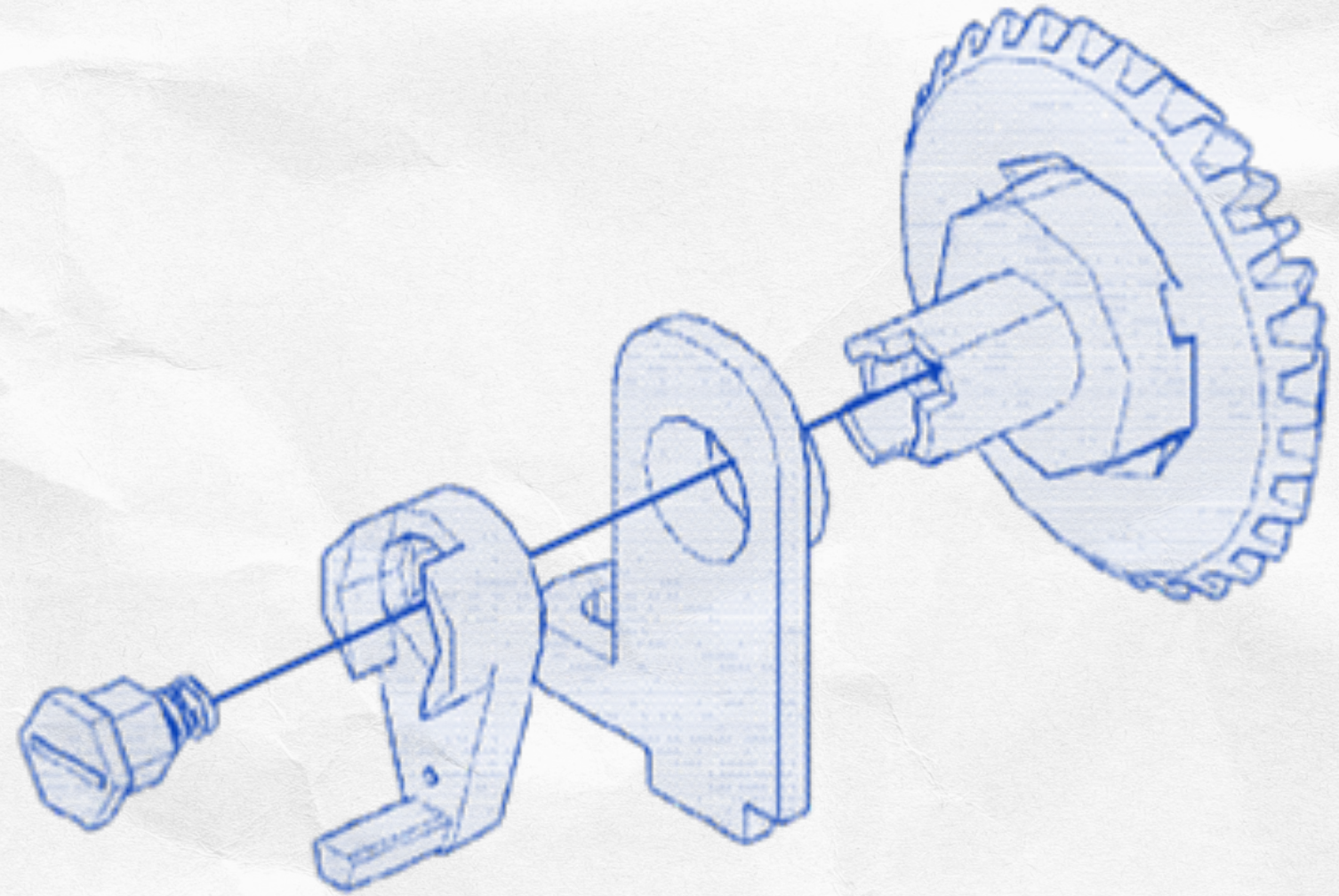


Let's ASSEMBle it!

2

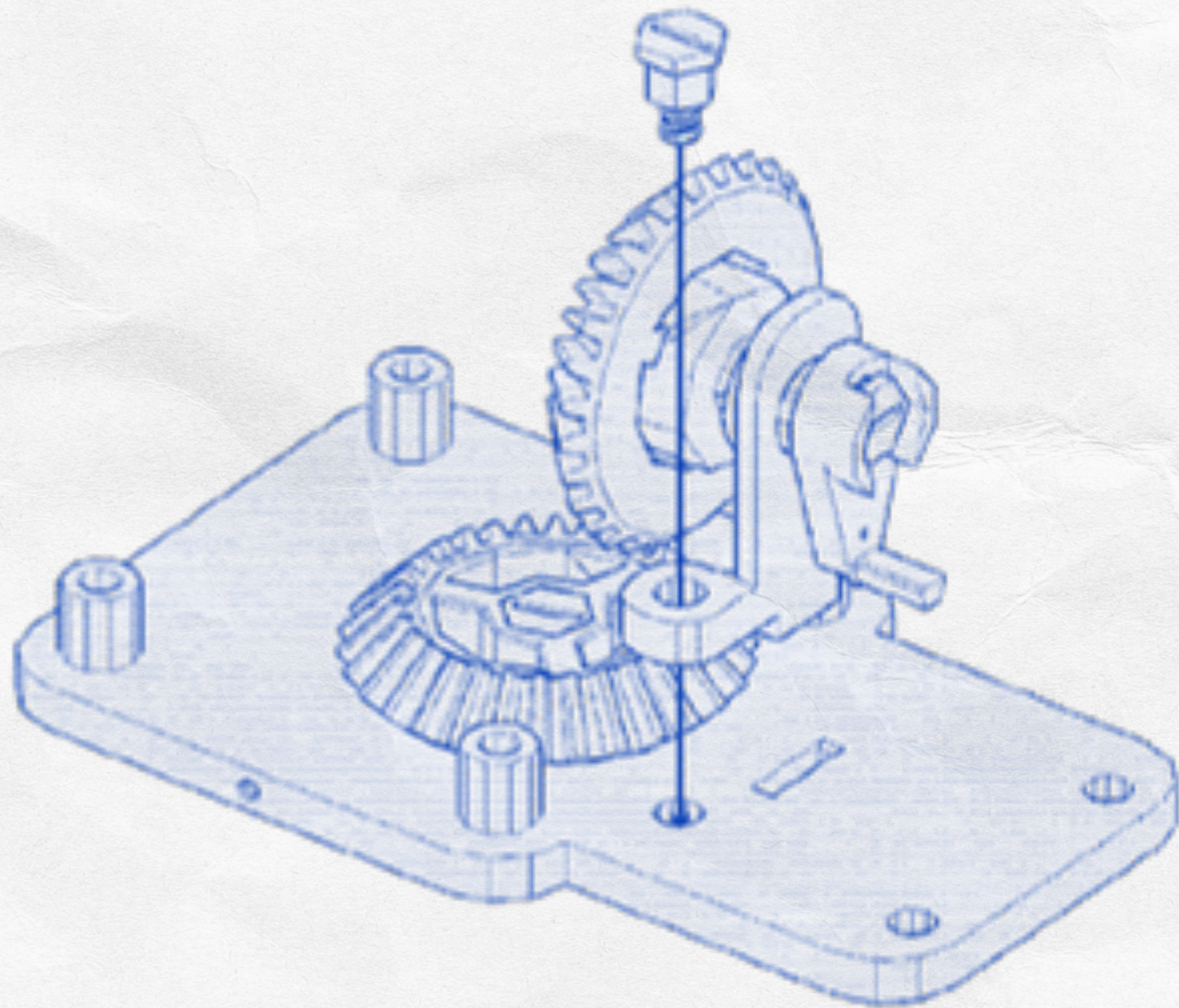


3

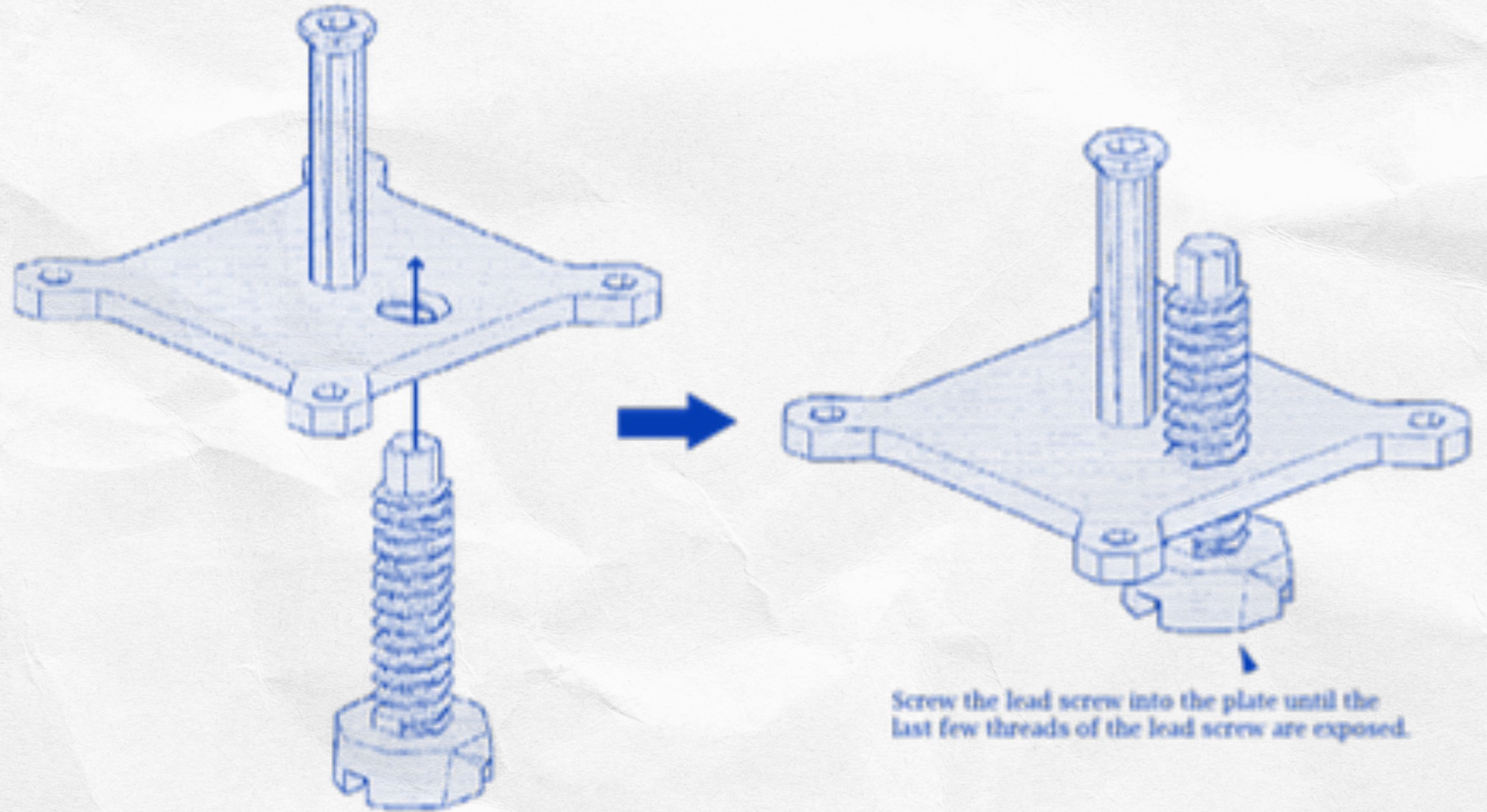


Let's ASSEMBle it!

4



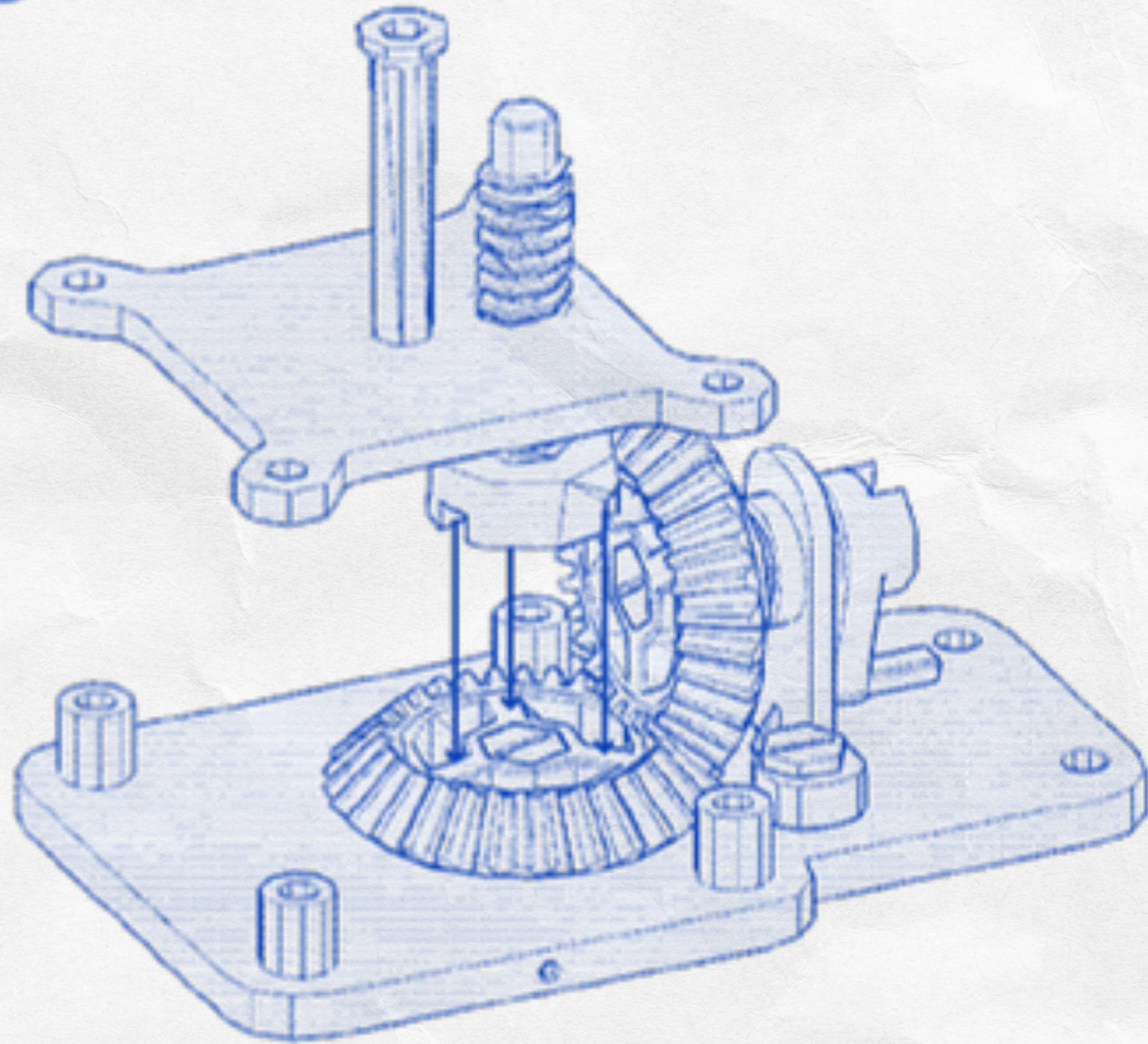
5



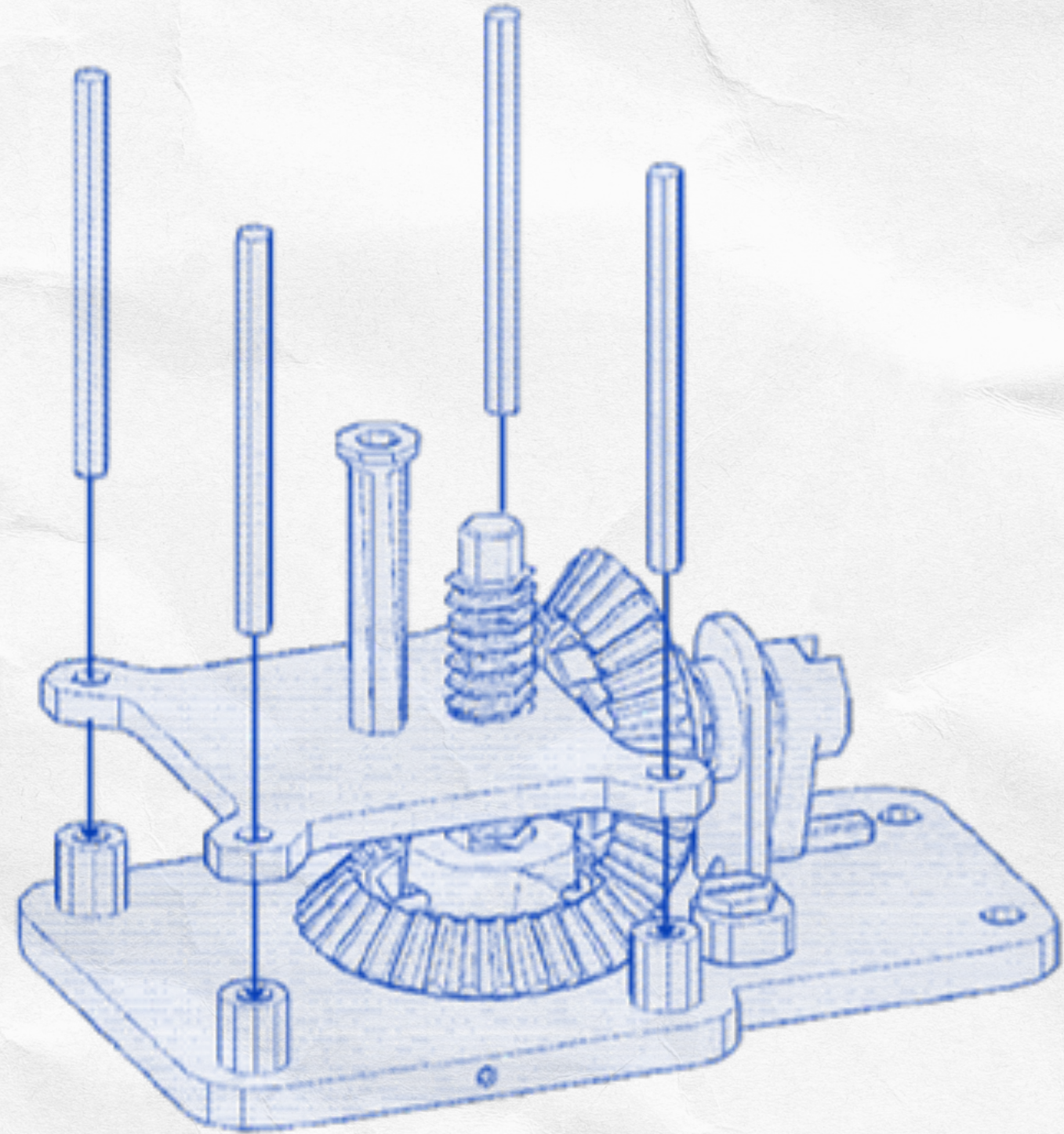
Screw the lead screw into the plate until the last few threads of the lead screw are exposed.

Let's ASSEMBLE it!

6

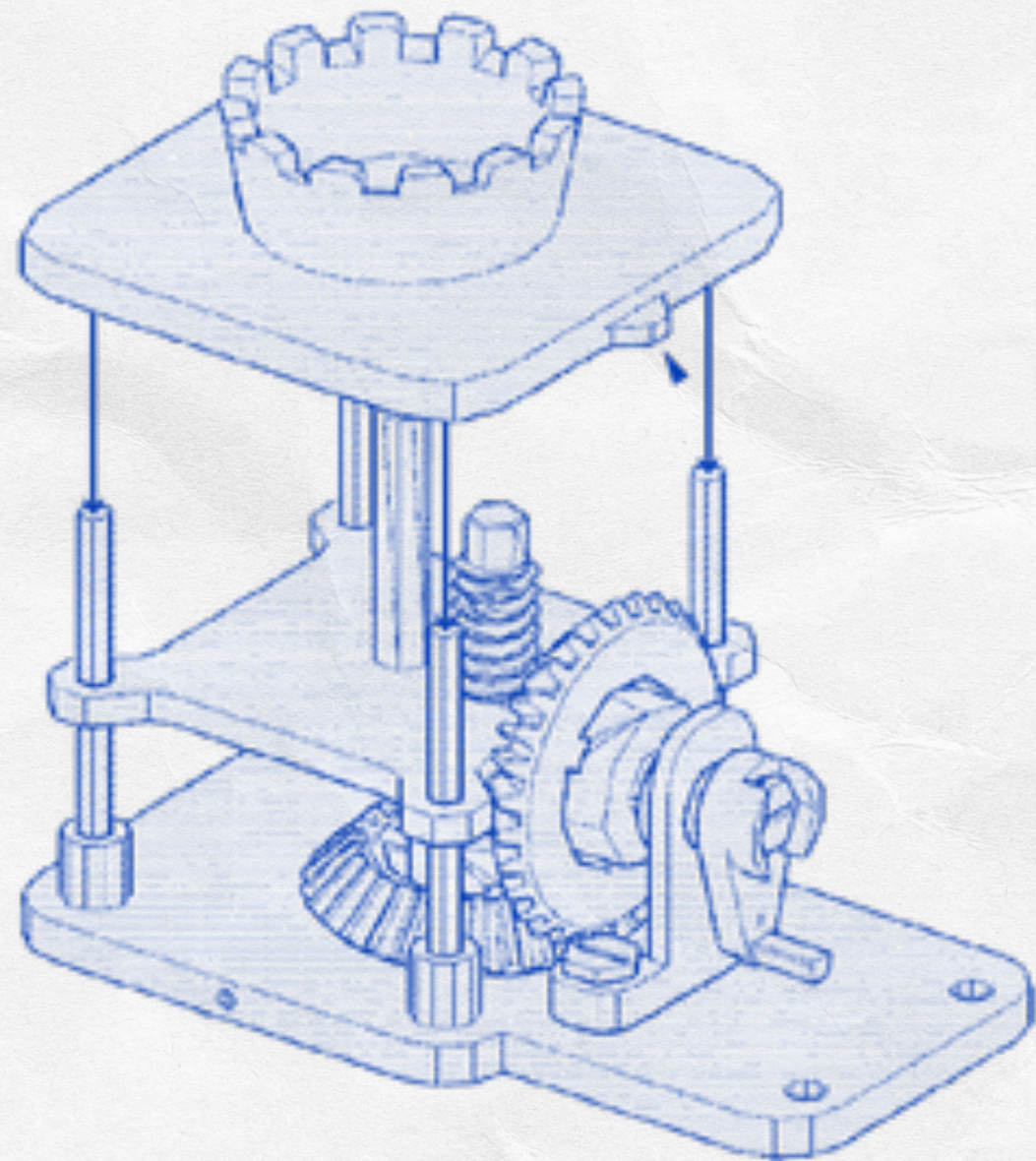


7

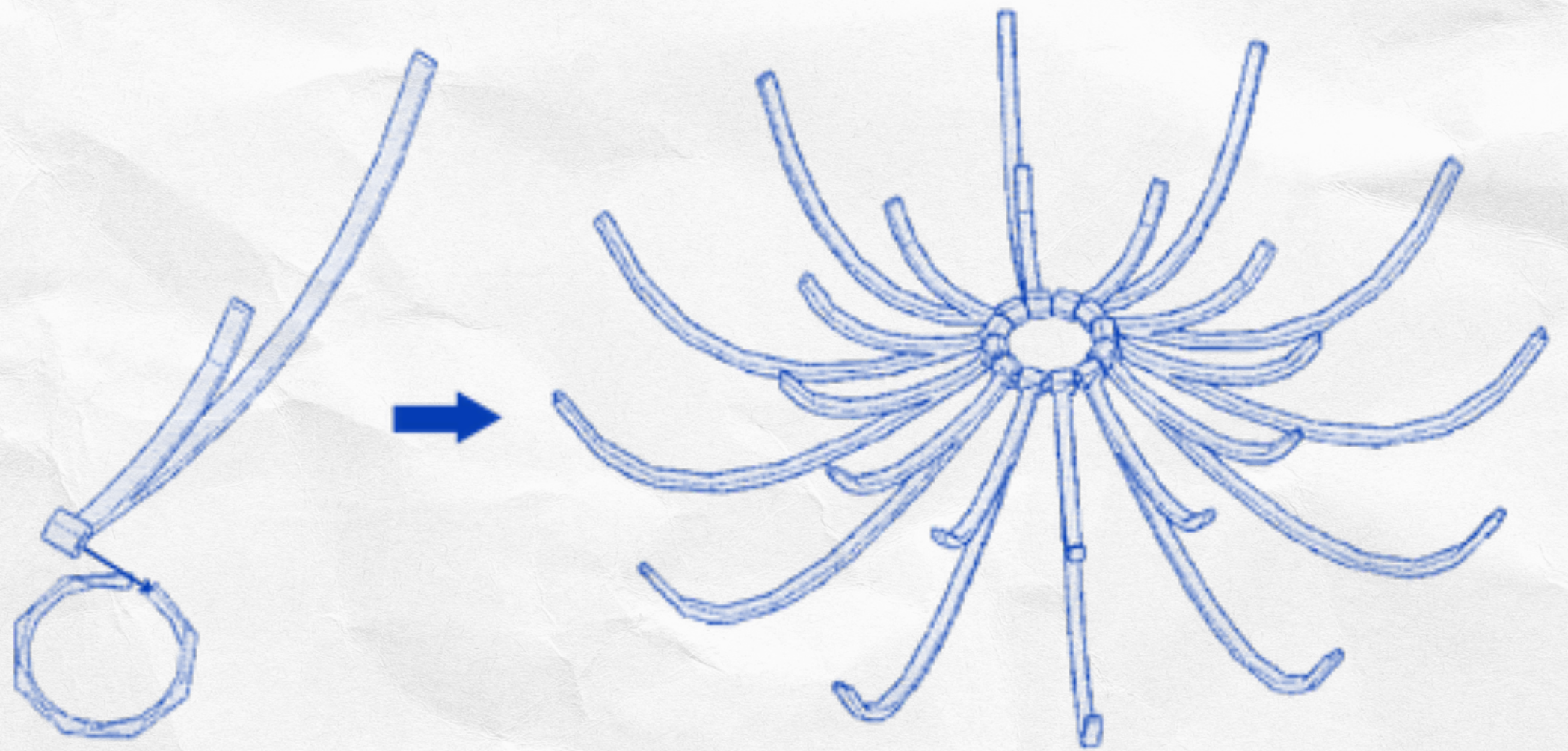


Let's ASSEMBle it!

8



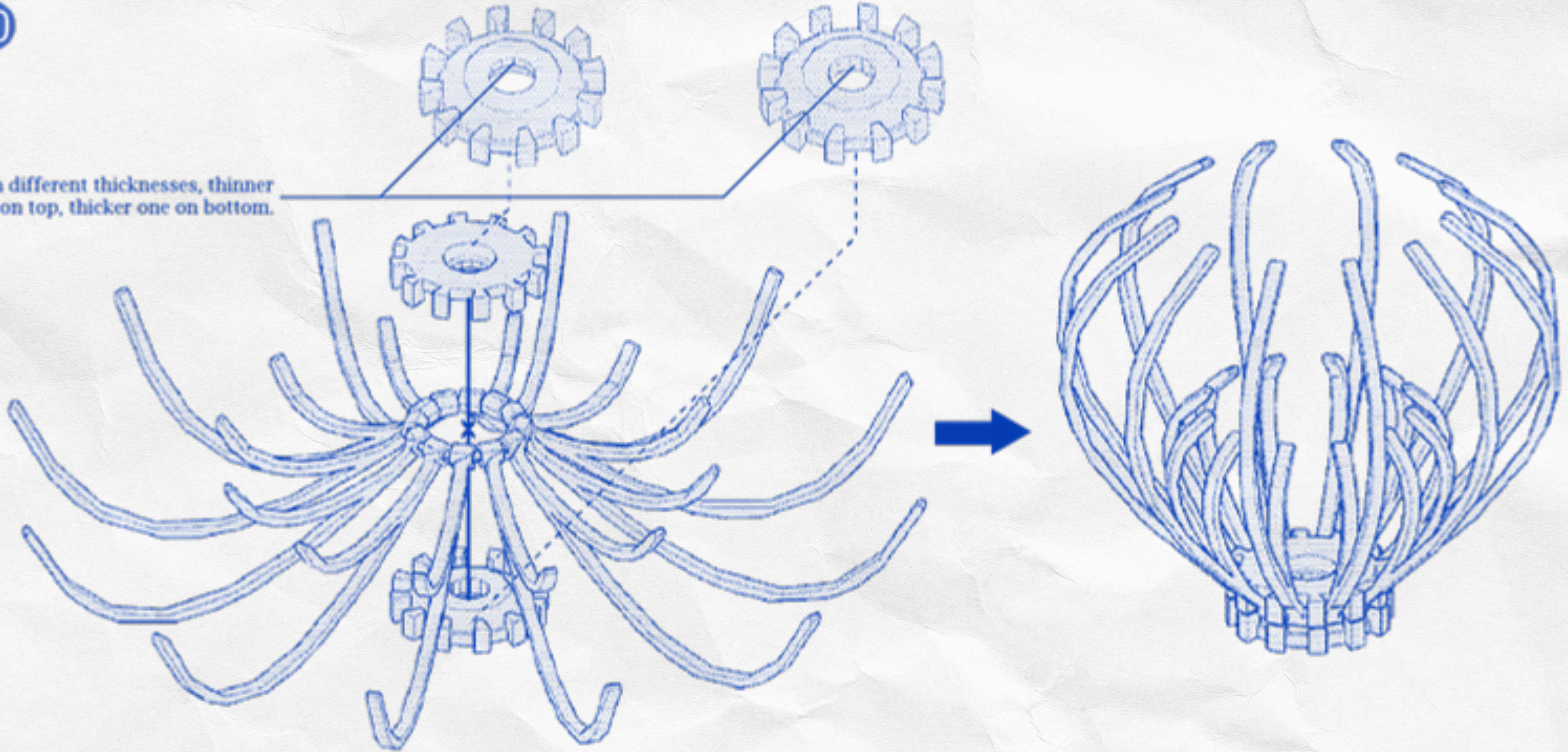
9



Let's ASSEMBle it!

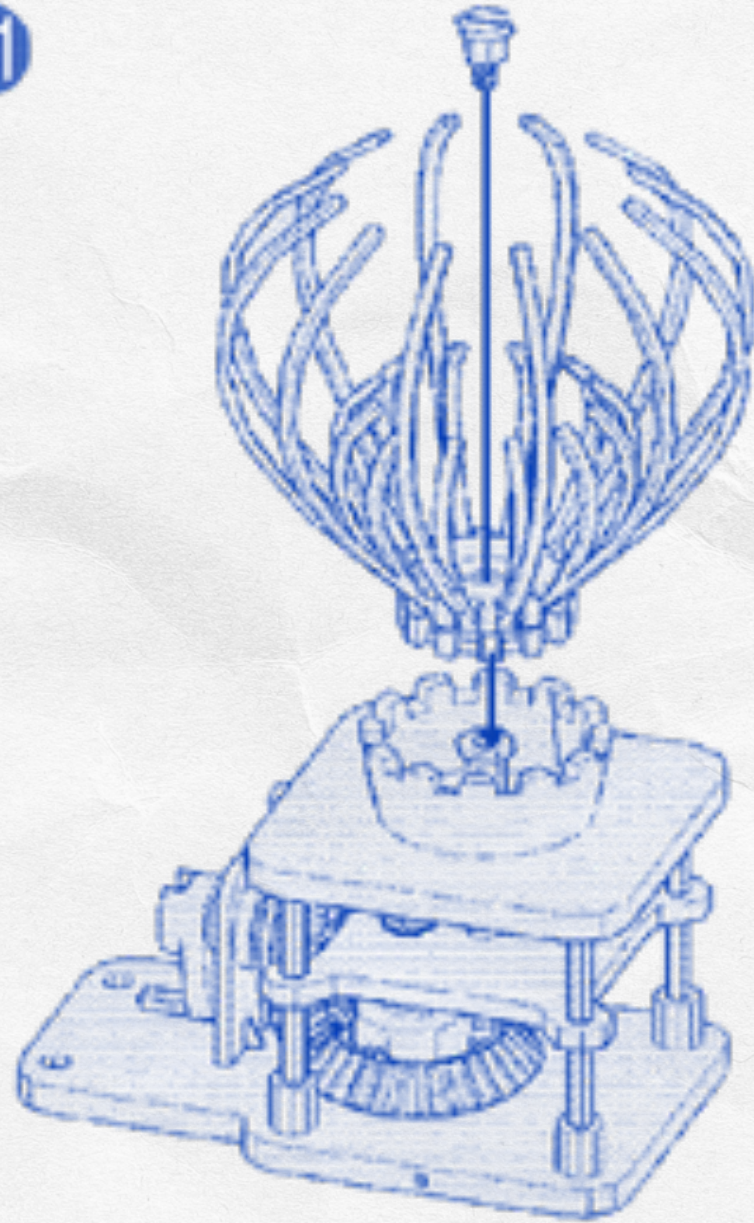
10

With different thicknesses, thinner one on top, thicker one on bottom.



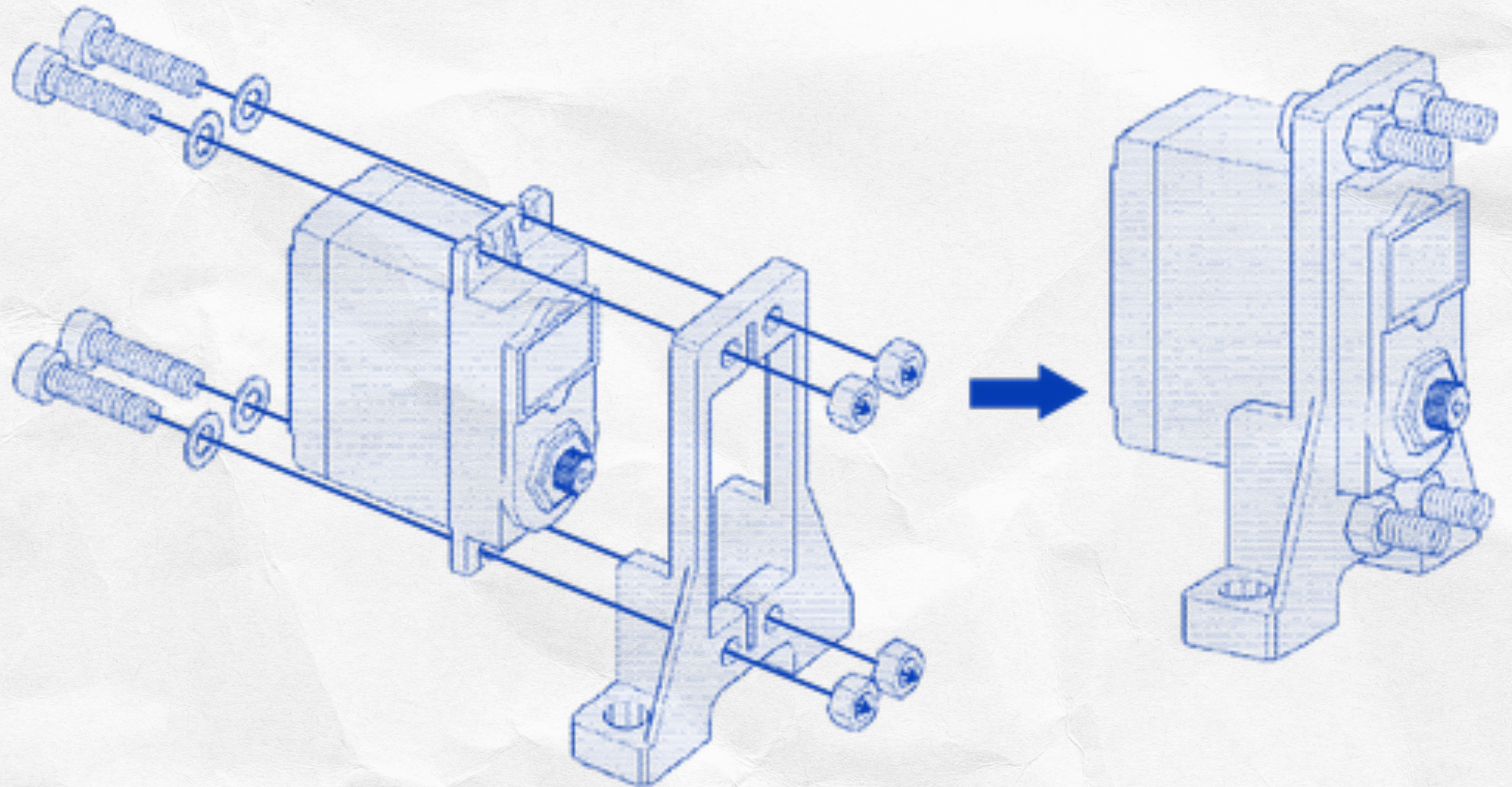
Let's ASSEMBLE it!

11



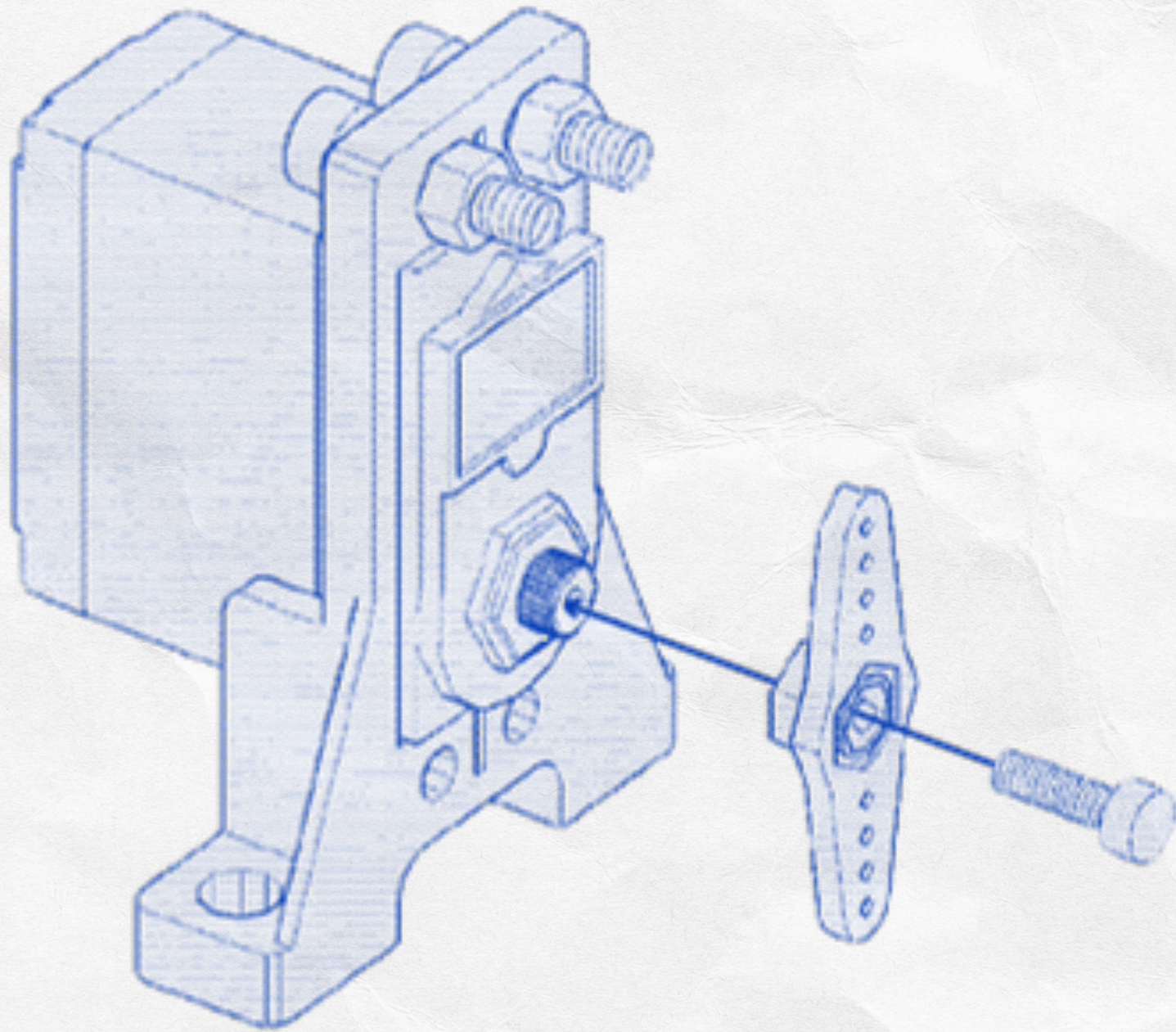
Now you can manually rotate to make the petals move.

12

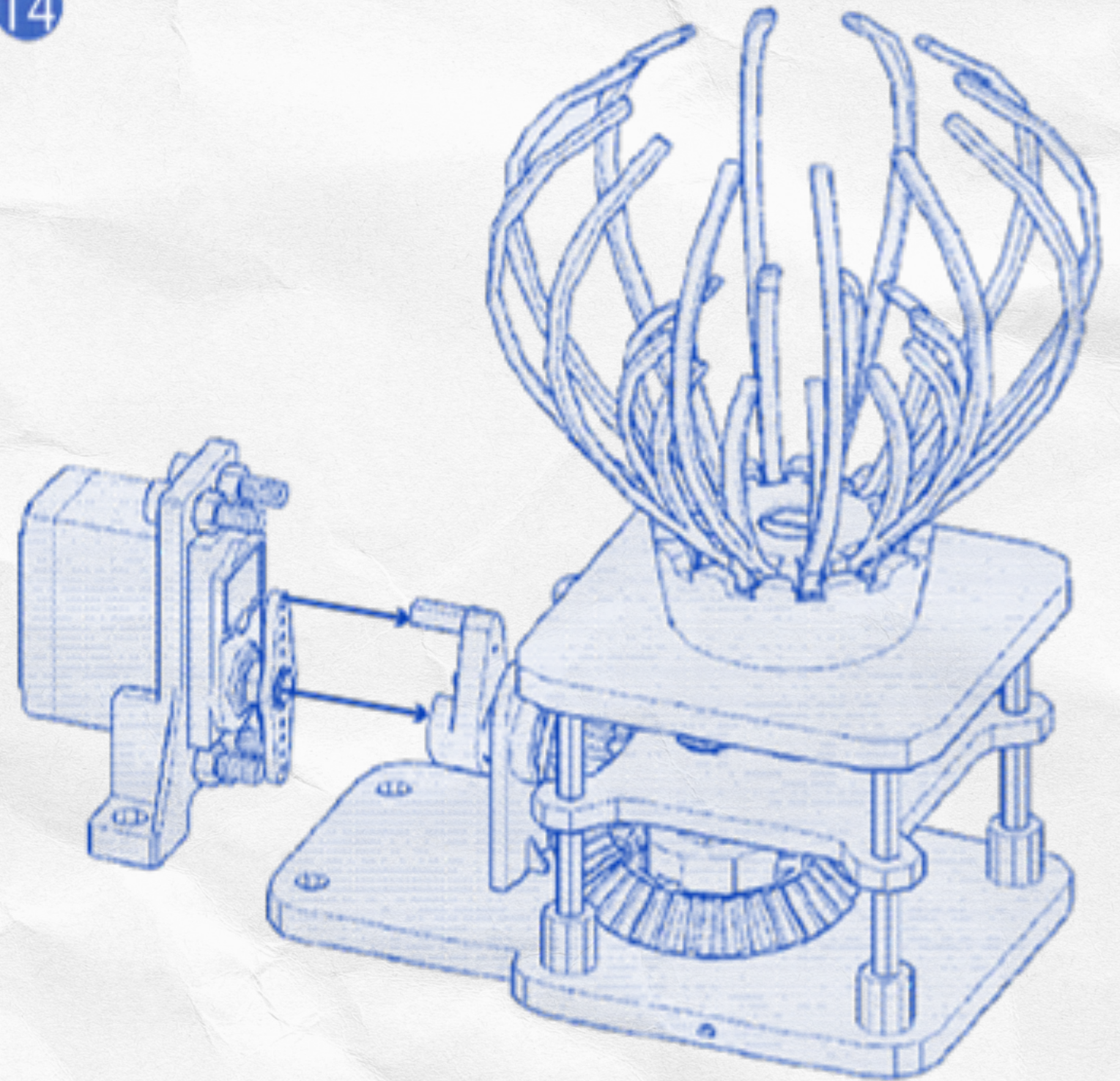


Let's ASSEMBLE it!

13

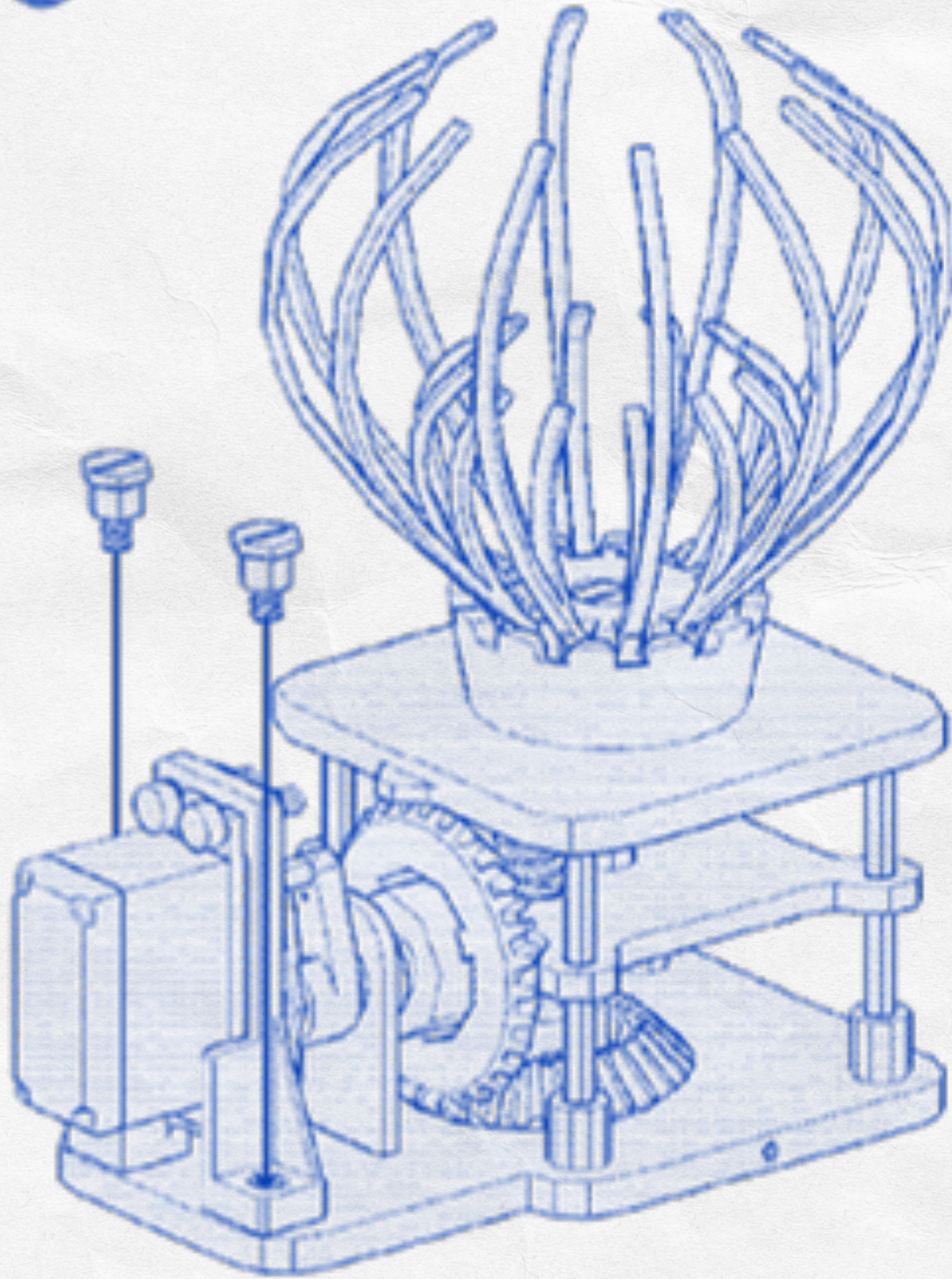


14

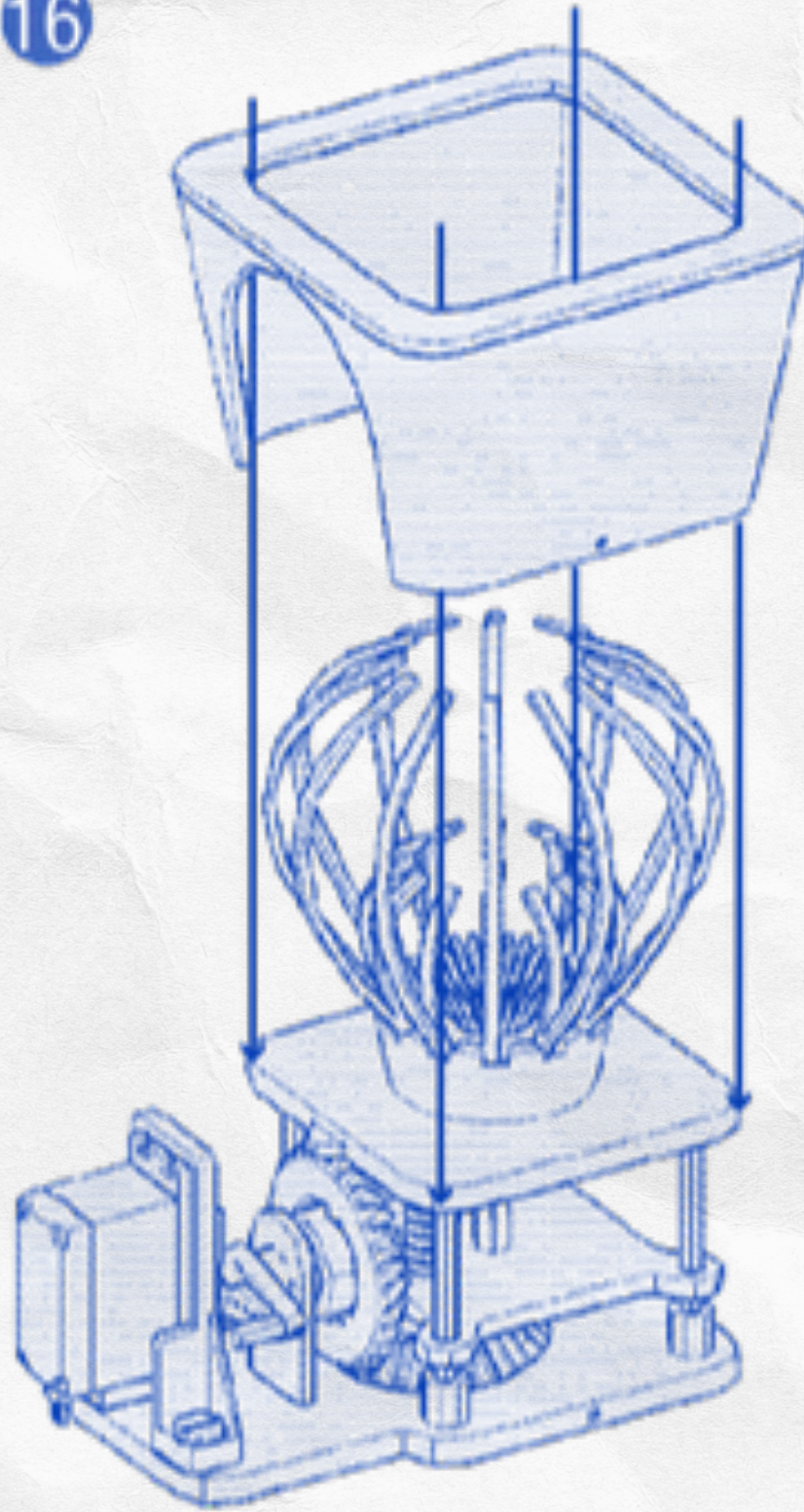


Let's ASSEMBLE it!

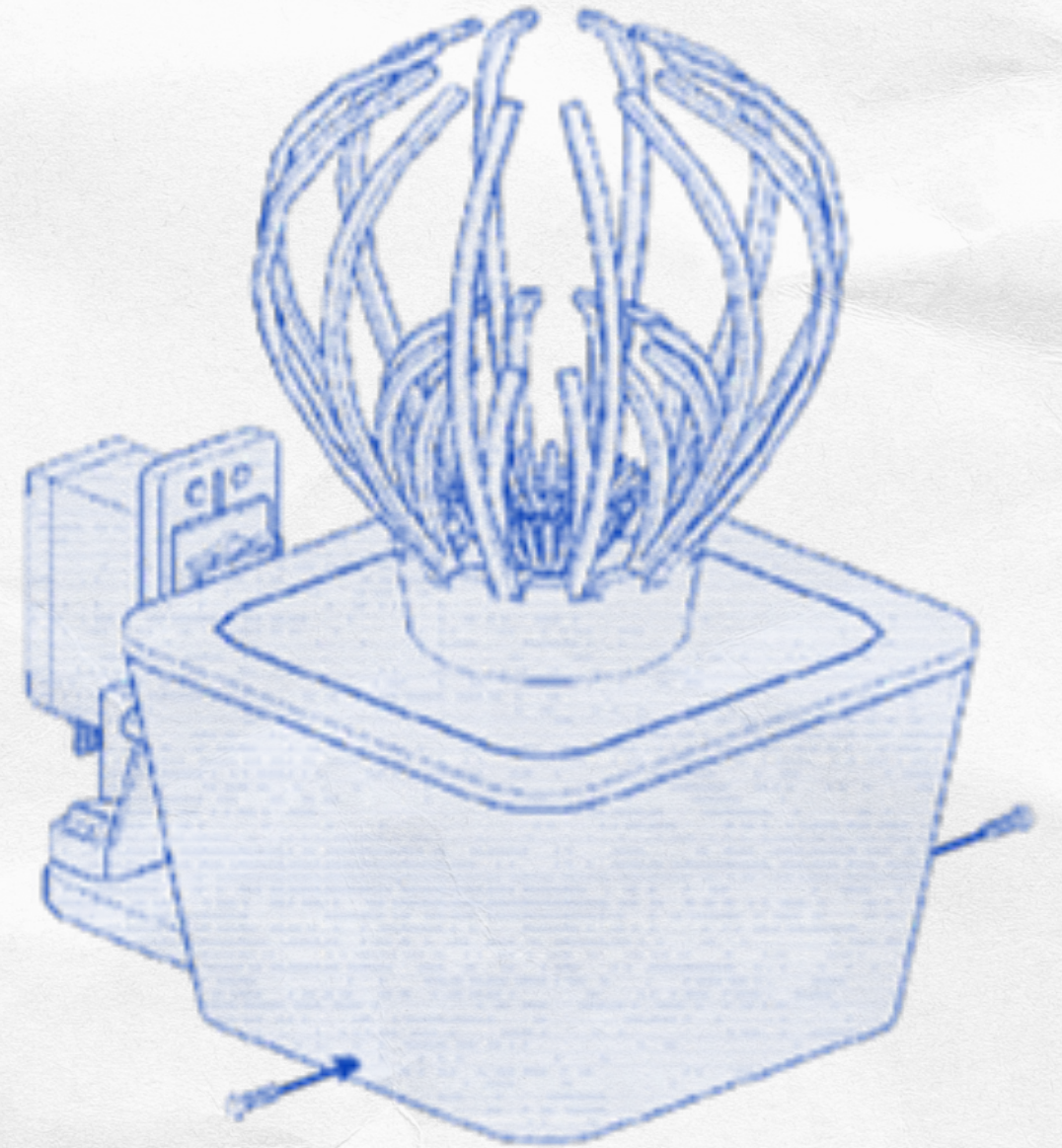
15



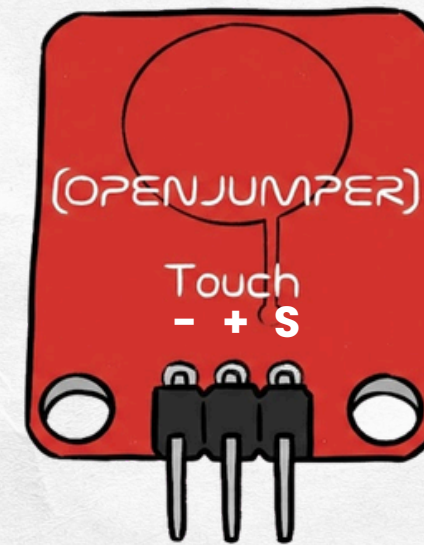
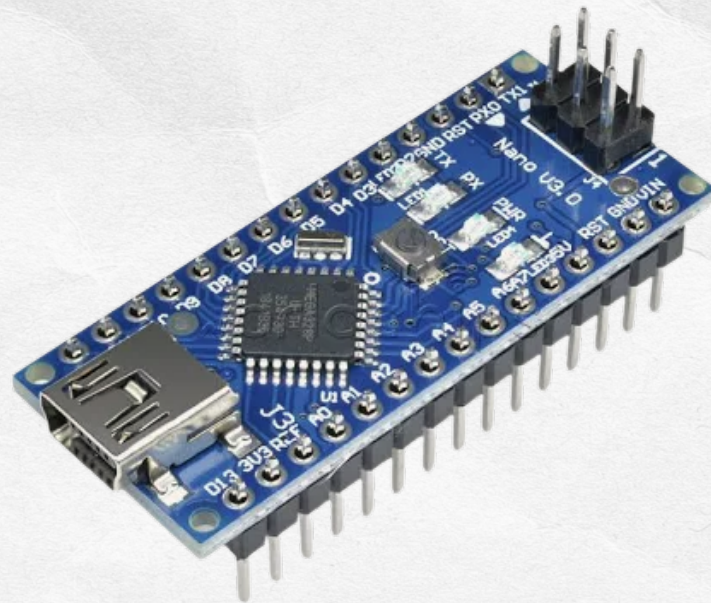
16



17



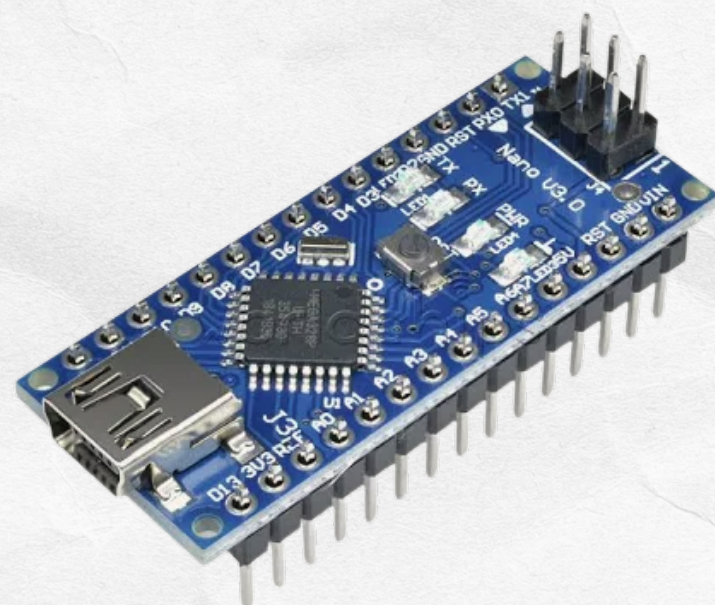
"Brain" of Mechanical flower



Development Board: **Arduino Nano** Touch Sensor



What do you need to prepare?



Arduino Nano



Data cable

(Must support data transmission;
do not use cables that only charge)



Computer



Install Arduino IDE



Arduino IDE (Integrated Development Environment)
Default Page





Development Enviroment Configuration

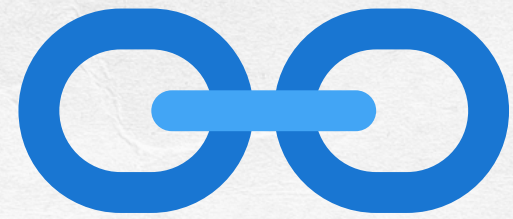
Update Board Manager

1. Click 'BOARDS MANAGER'



2. Search 'AVR'

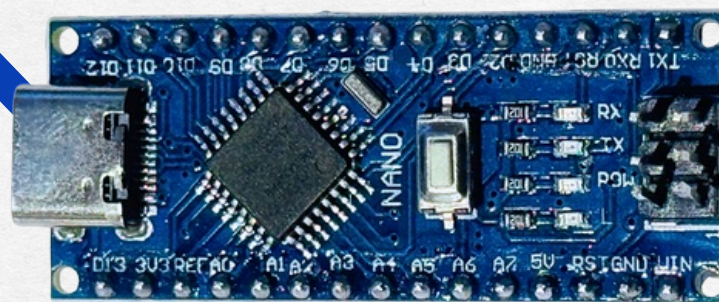
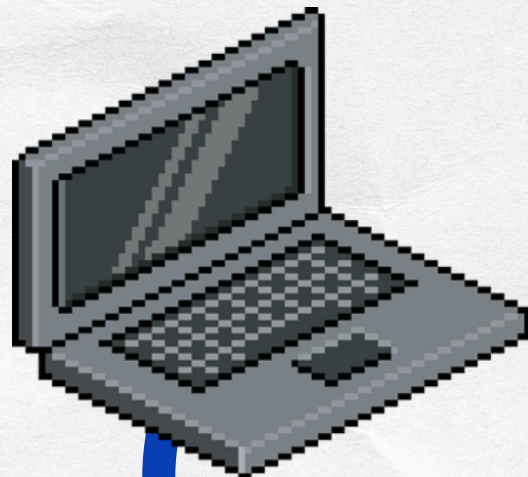
3. Update 'Arduino AVR Boards'



Let's **Test** Your Laptop Communicate with Arduino Nano Successfully !

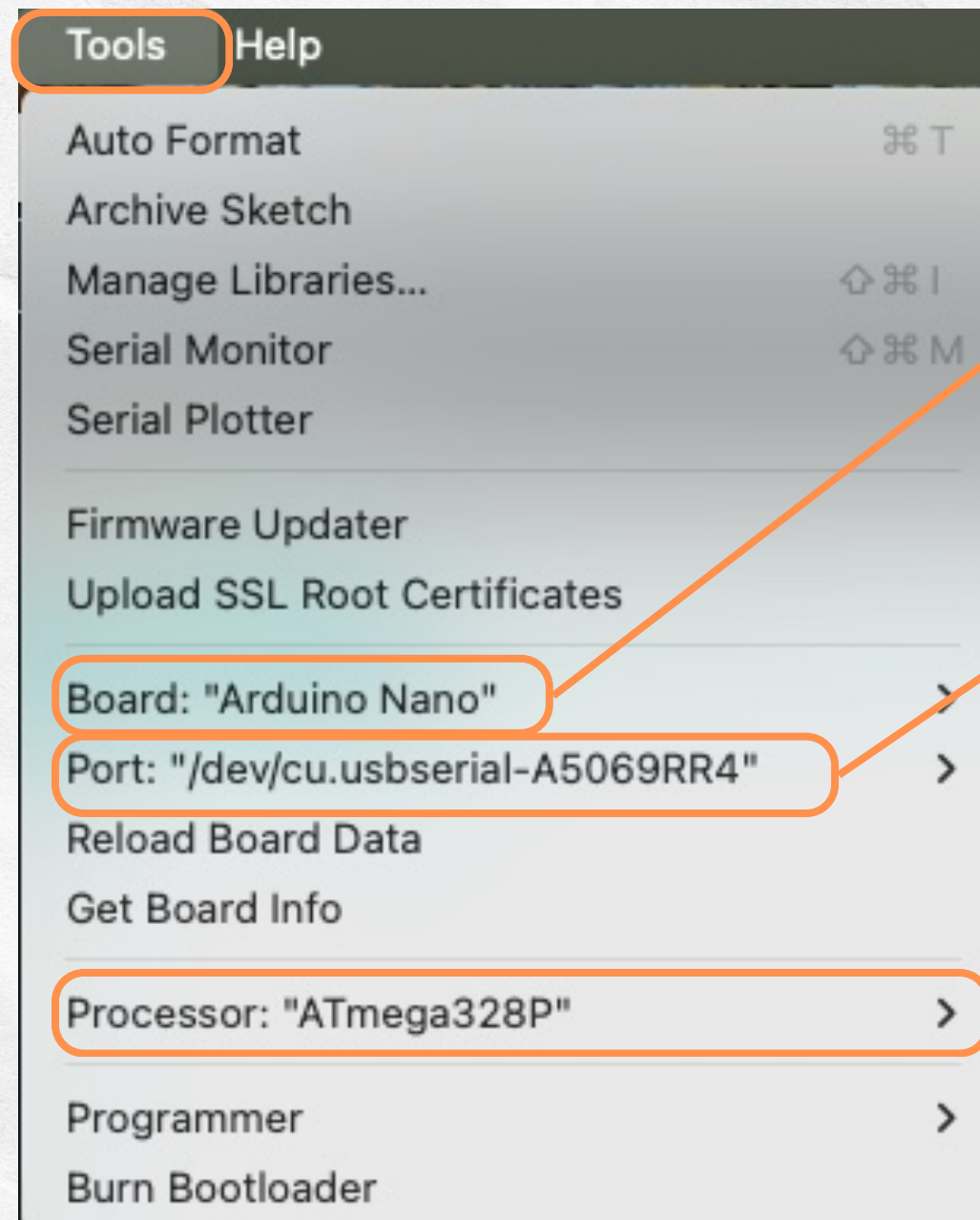
1

Connect the Laptop with Arduino Nano



2

Select respective Board, Port, Processor



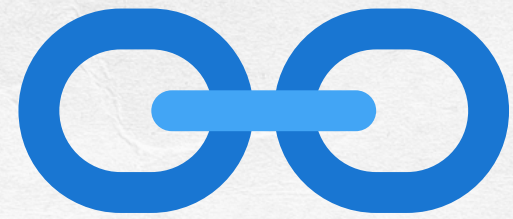
1. Select 'Arduino Nano'

2. Select newest port

Windows always is COMx;
Mac is /dev/cu.usbserial-xxx)

3. Select 'ATmega328P'





Let **Test** Your Laptop Communicate with Arduino Nano Successfully !

3

Use simple code to test the communication



Open the Github link

Link: <https://shorturl.at/78yy2>

Please copy the 'Nano_TEST_EASY' code



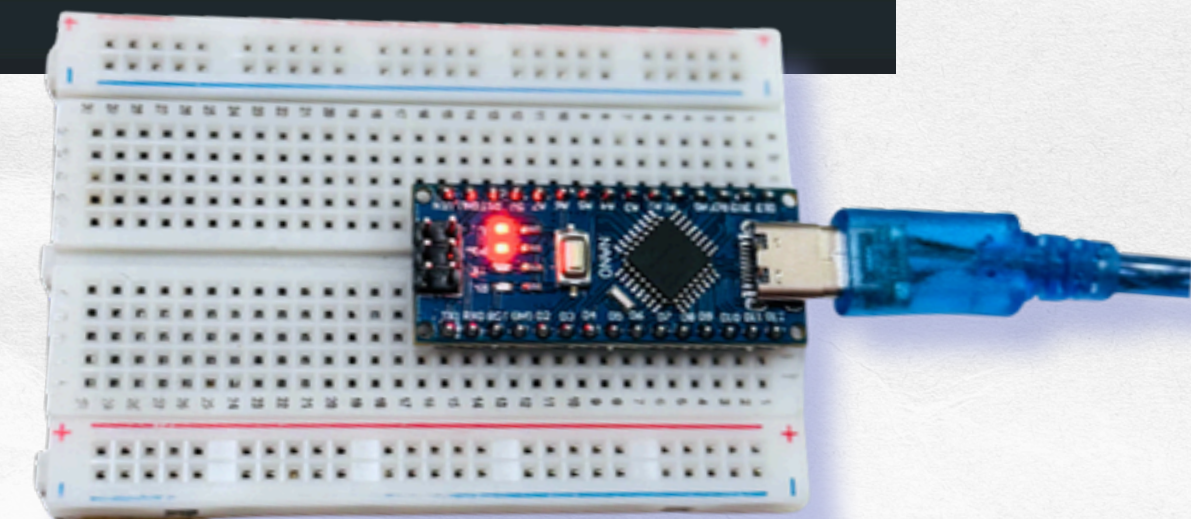
NANO_TEST_EASY



1.Verify

2.Upload

```
1 //Validation Communication
2
3 void setup() {
4   pinMode(LED_BUILTIN, OUTPUT);
5
6 }
7
8
9 void loop() {
10
11   digitalWrite(LED_BUILTIN, HIGH);
12
13 }
```



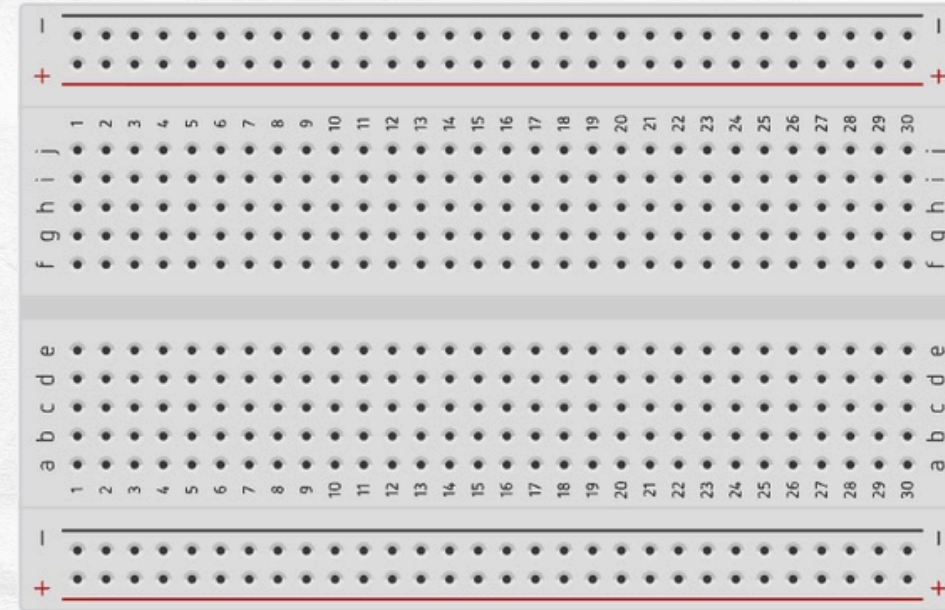


Part 1 – Machine Flower Control by Buttons

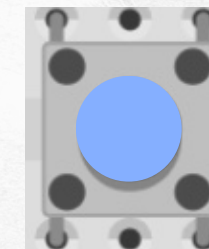
Checklist



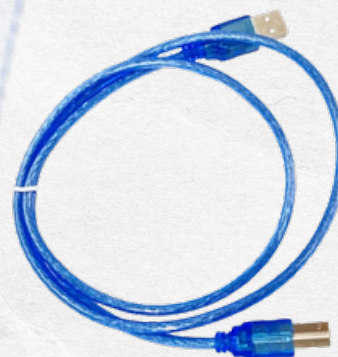
Arduino Nano



Bread Board



Button x 2



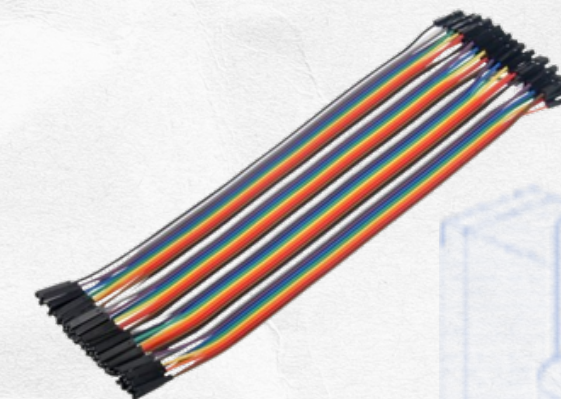
Data Cable



Servo MG996R



Battery

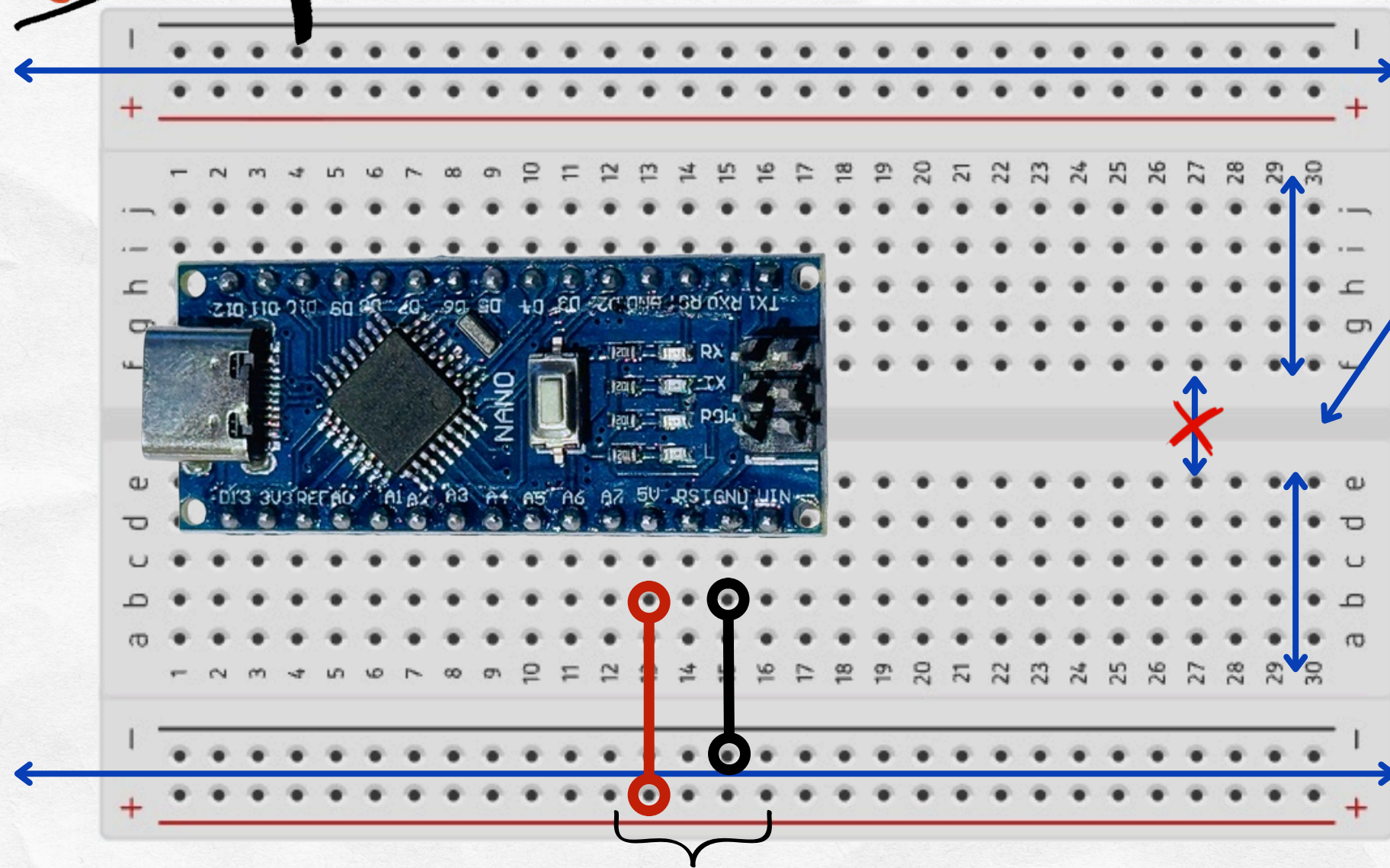


Dupont Lines



Bread Board Connection

Solderless Board



Power Rails (Horizontally Connected),
1 through 30 fully linked, no connection between each other rail

Center Channel

Terminal Strips, Each column is vertically connected,
no connection between each strip, between e and f

Power Rails (Horizontally Connected),
1 through 30 fully linked, no connection between the two rails

Export **+5V** and **GND** on Bread Board

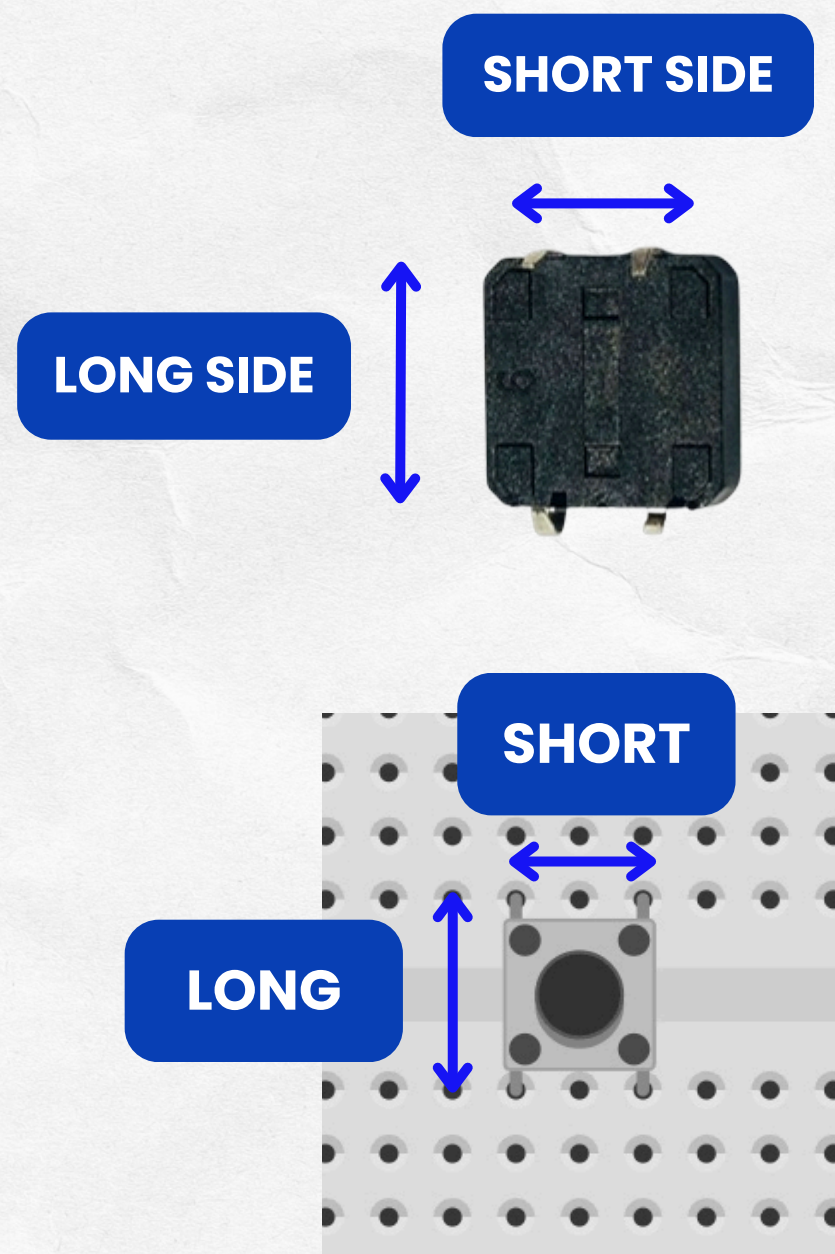
Let's go button connection!



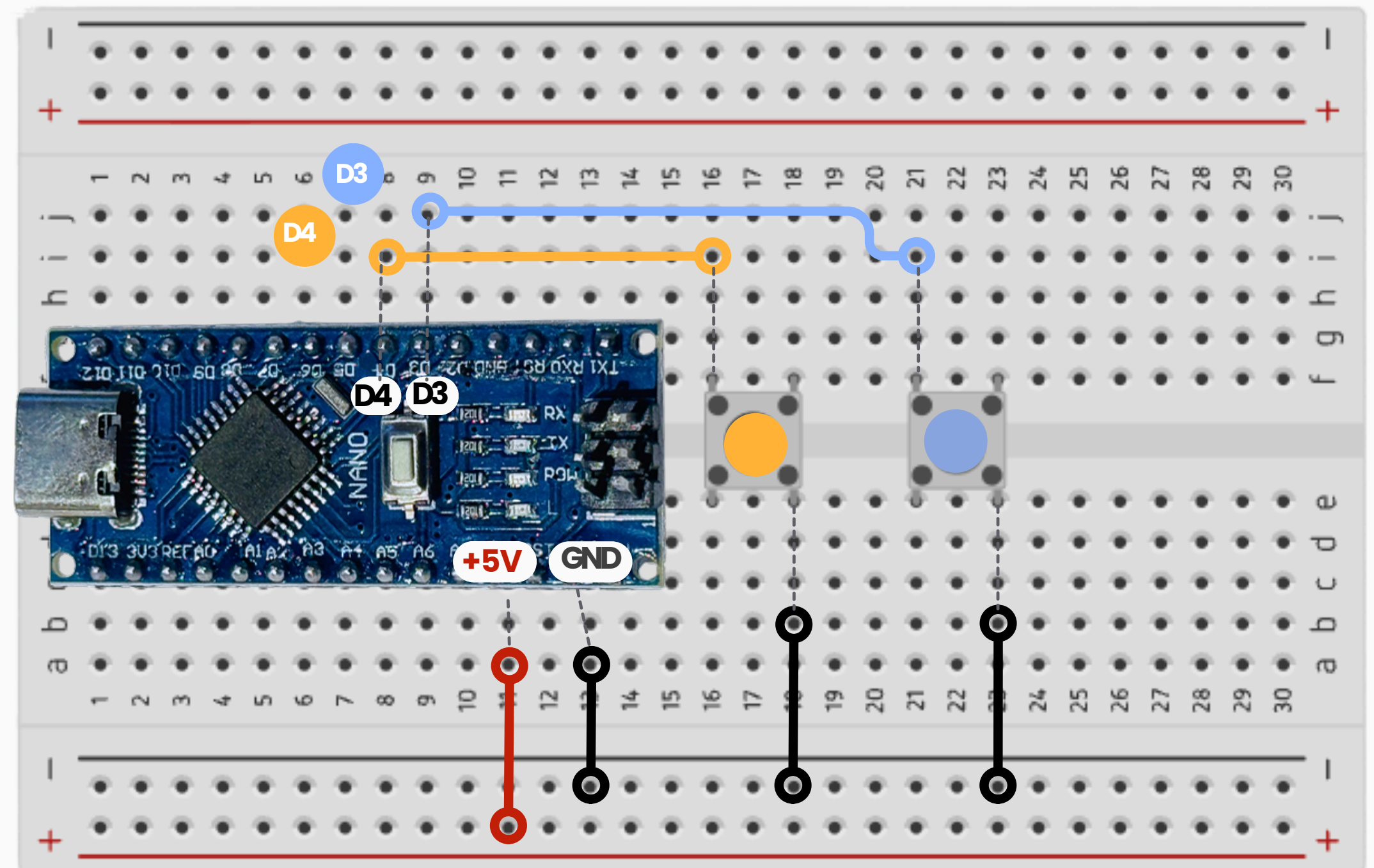
BUTTON

Reading Button State (0 or 1)

1 Button Wiring & Pin Connection



6 lines in total





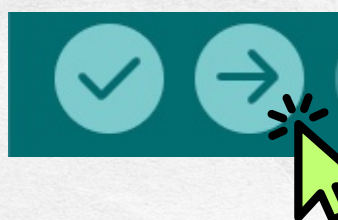
BUTTON

Reading Button State (0 or 1)

2

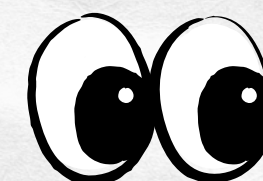
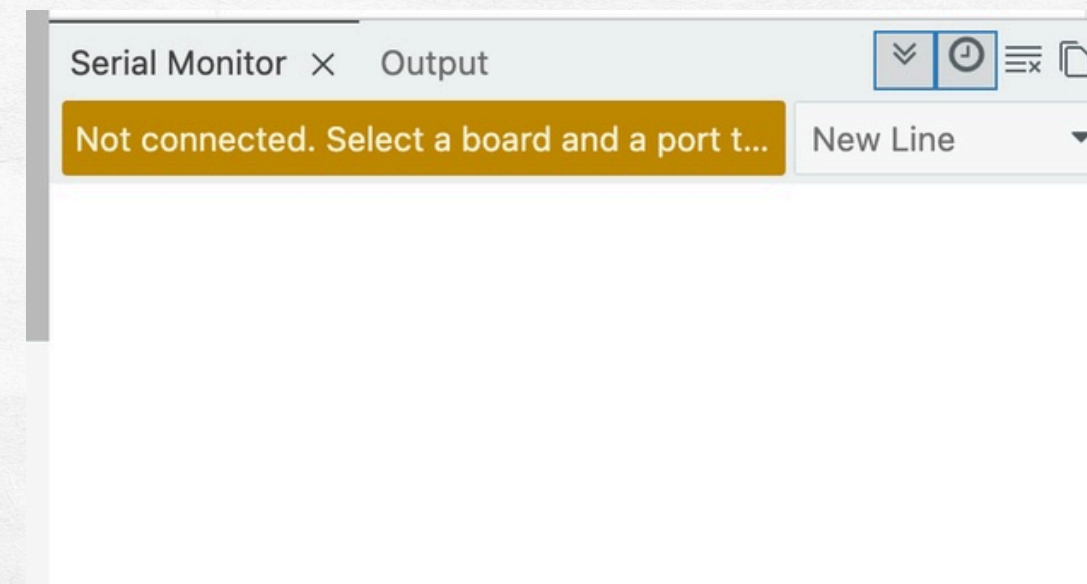
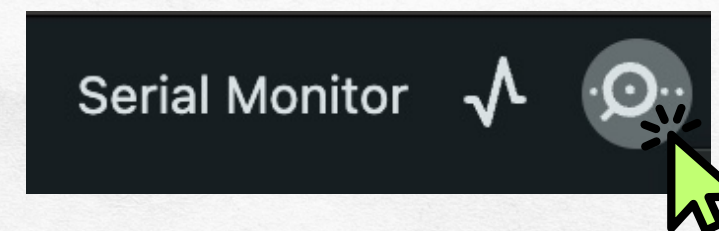
Please copy the '**MF_Button_READVALUE**'

Connect **Nano** with **Laptop** by data cable



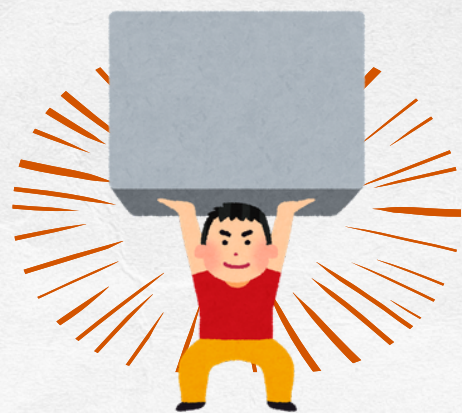
Upload the code to Nano

```
MF_Button_READVALUE.ino
1  // Read Button State
2
3  int buttonPin = 3; // D3
4
5  void setup() {
6
7      Serial.begin(9600);
8      // INPUT_PULLUP, the pin is HIGH by default.
9      pinMode(buttonPin, INPUT_PULLUP);
10
11  }
12
13  void loop() {
14      // Read the state of the digital pin
15      // 2 Results: 1 (HIGH) or 0 (LOW)
16      int buttonState = digitalRead(buttonPin);
17
18      Serial.println(buttonState);
19      delay(200);
20  }
```



Try to use press the button and see what happen on your Serial Monitor

Disconnect the Data Cable,
Let's go Servo connection!



Let's make the flower move with the Servo!

MG996R (360°) Servo

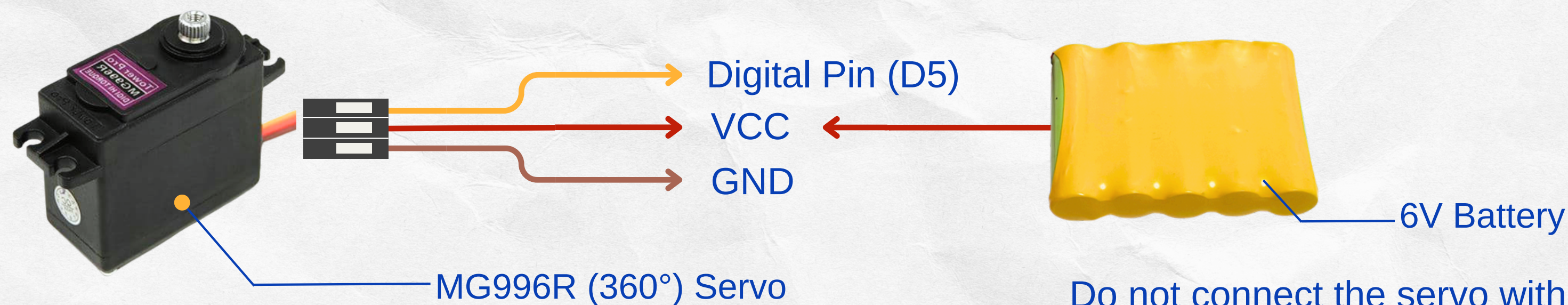


180° Servo

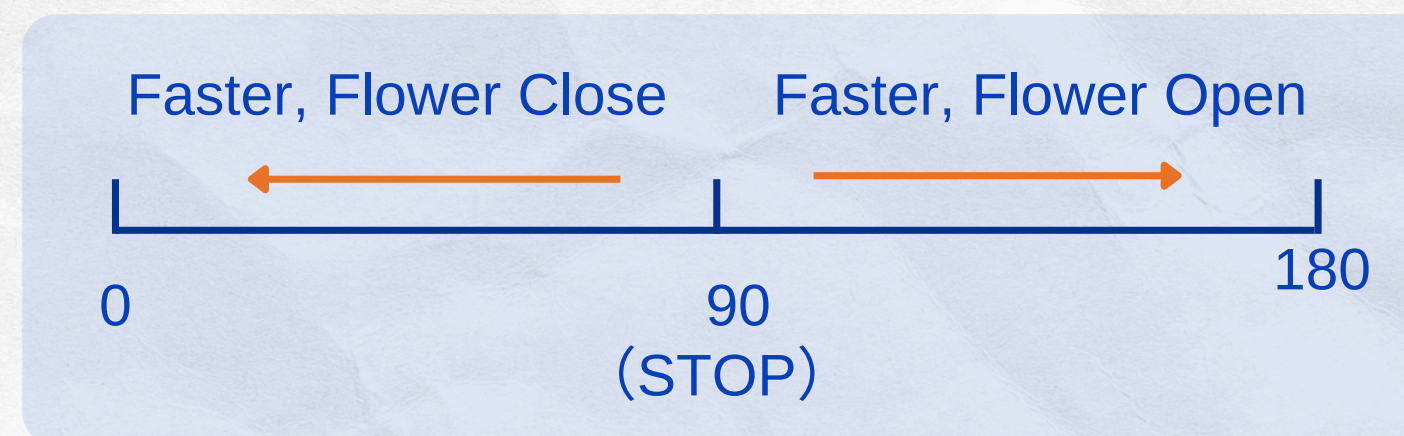
Controls rotation to a specific angle, max 180°

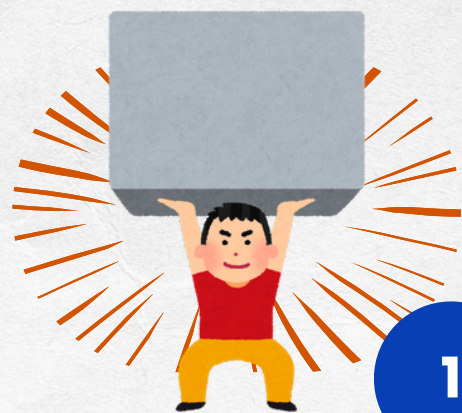
360° Continuous Servo

It is no longer about "rotating to a certain degree"
Control the rotation direction and rotational speed.



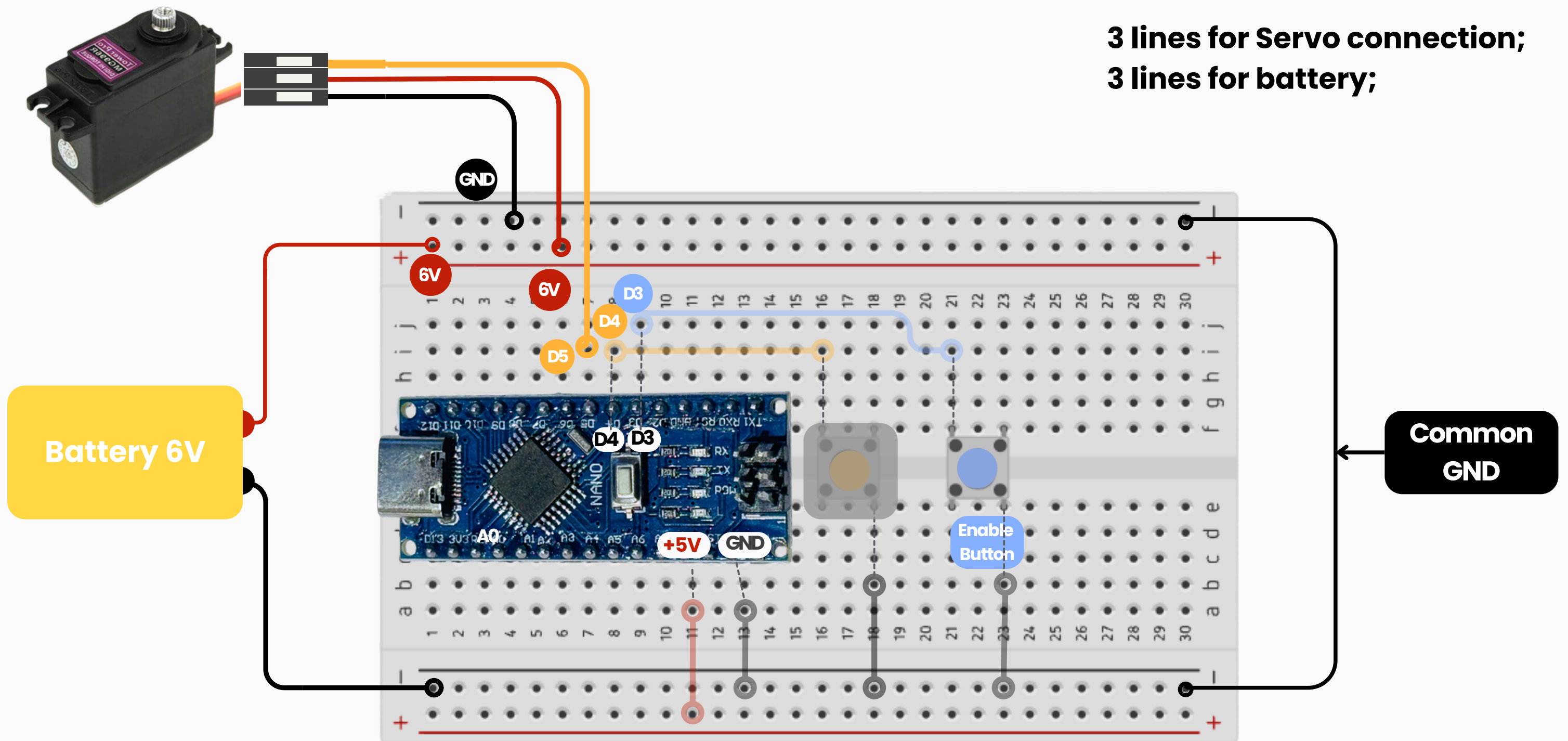
Do not connect the servo with 5V pin on Nano

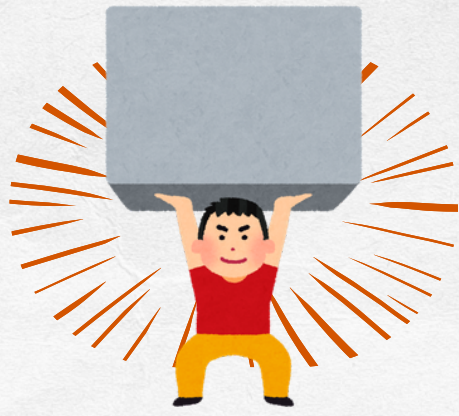




Let's make the flower move with the Servo!

1 Follow the diagram to connect:



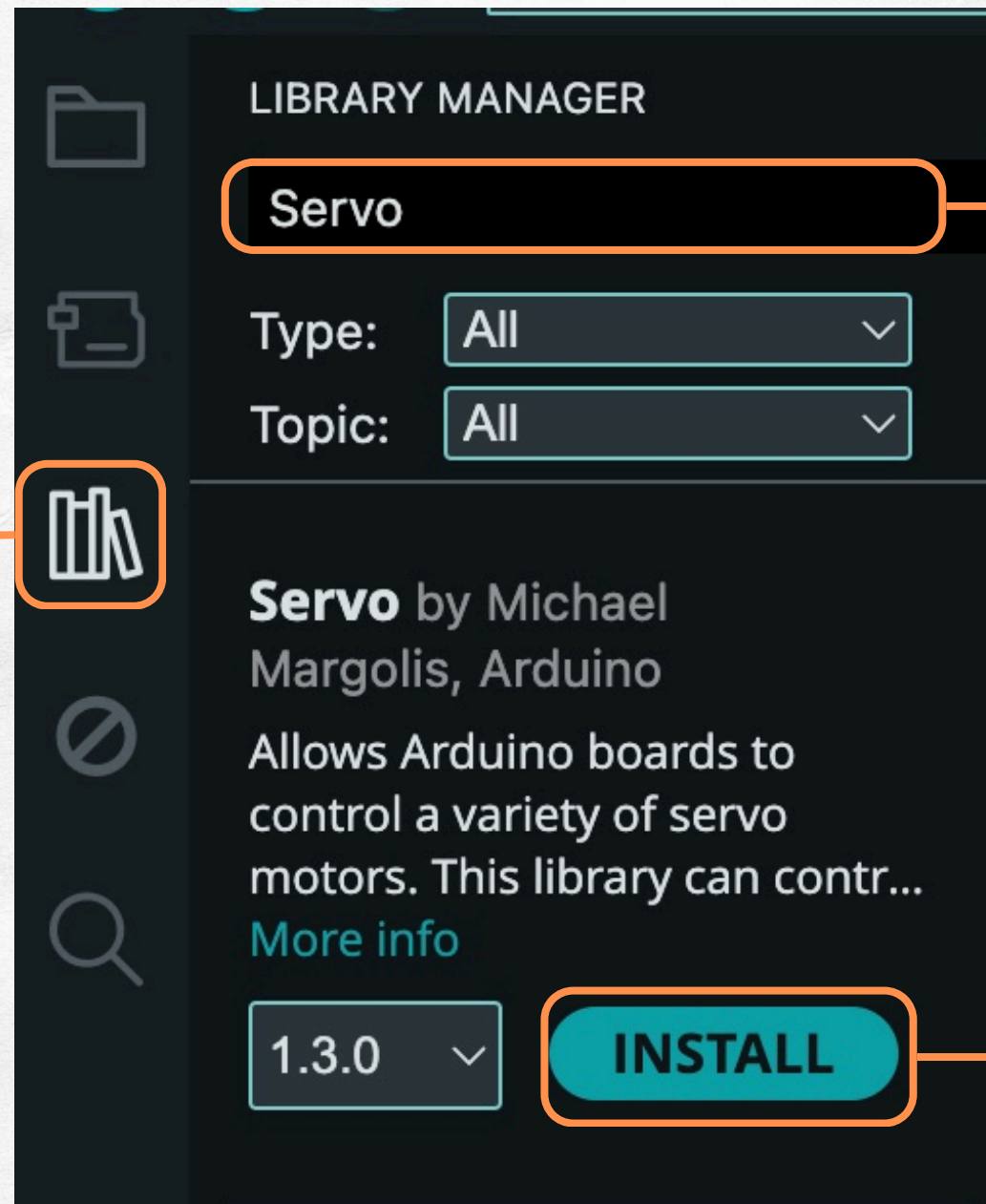


Let's make the flower move with the **Servo**!

2

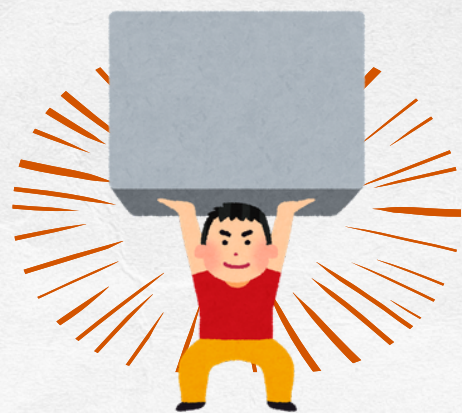
Install Library

1. Click 'Library'



2. Search 'Servo'

3. Install 'Servo Library'



Let's make the flower move with the Servo!

3 Please copy the 'MF_Button_RESET' code

MF_Button_RESET

```
/*
Mechanical Flower - Button Reset
- Press (D3) to Rotate Right or Left.
- Double Press (D3) to Stop.
- After Stop, default start to Open the flower.
- Rotate Right, Flower Open; Rotate Left, Flower Close.
*/
#include <Servo.h>

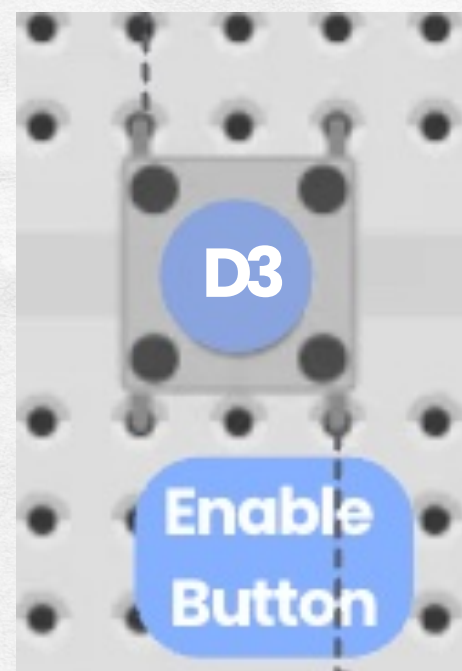
Servo servo;

//Parameter Definitions
const int servoPin = 5;
const int buttonPin = 3;

const int SERVO_STOP_DEG = 90; // Stopped
const int SERVO_OPEN_DEG = 135; // Rotate Right / Move Up (Opening speed)
const int SERVO_CLOSE_DEG = 45; // Rotate Left / Move Down (Closing speed)

enum ServoState {
    ROTATING_RIGHT,
    ROTATING_LEFT,
    STOPPED
};

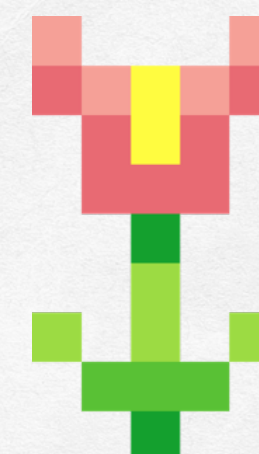
ServoState currentState = STOPPED; // Initial state set to STOPPED
```



- You can control the mechanical flower by press "**Enable Button D3**"



Press Once
Change Rotate Direction



Double Click
Rotate Stop

4 Connect **Nano** with **Laptop** by data cable;
Upload the code to Nano;

Before Disconnection, record the time it takes flower open from button to top.
Let's go Touch Sensor Control!

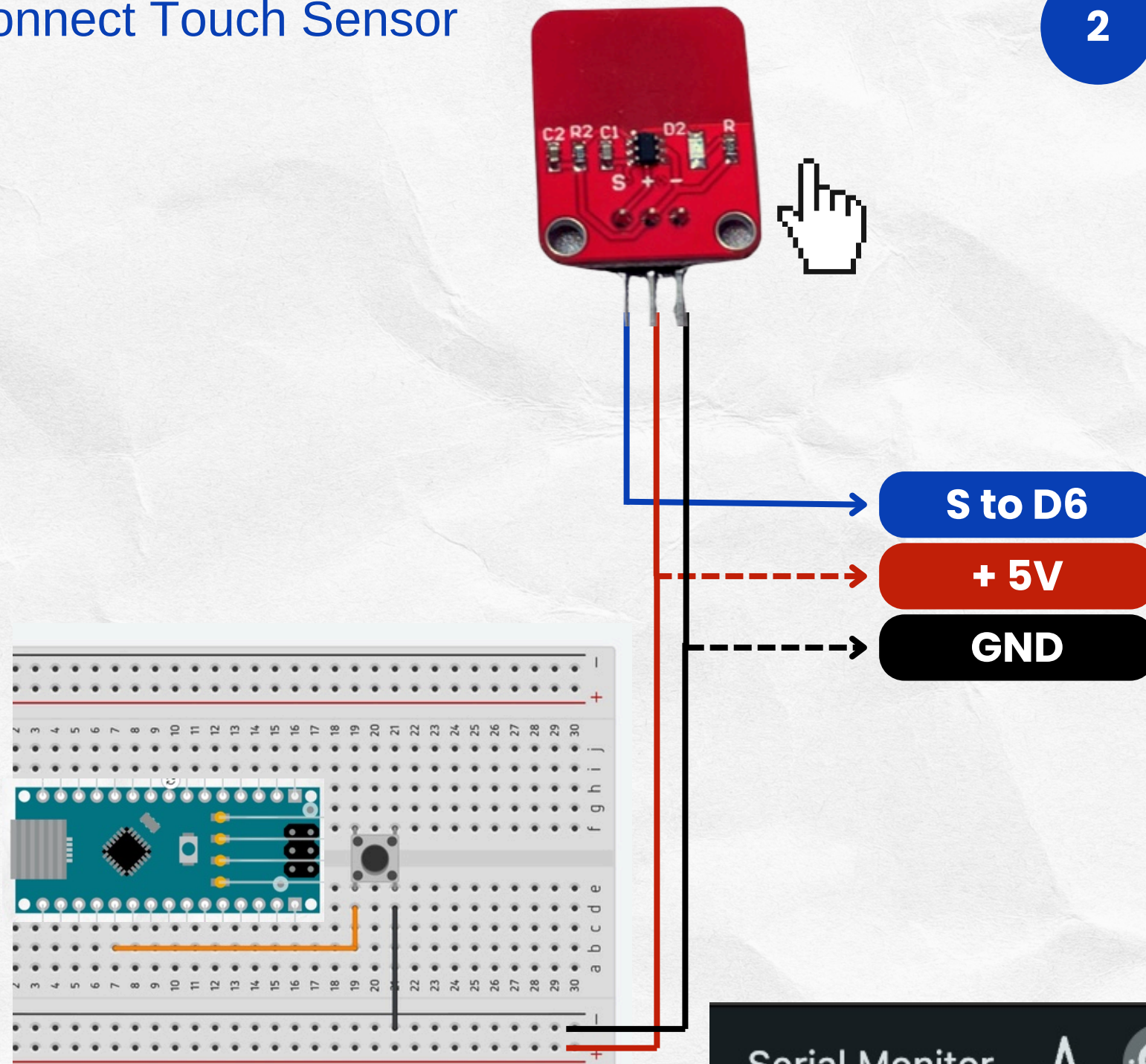


Part 2 – Let's Bloom Flower with Interaction!

Touch Sensor Control the Servo — Reading Sensor State

1

Connect Touch Sensor



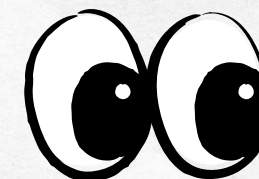
2

Connect **Nano** with **Laptop** by data cable;
Upload the '**MF_Touch_READVALUE**'

```
9  #include <Servo.h>
10 #include <EEPROM.h>
11
12 const int SERVO_PIN    = 5;
13 const int TOUCH_PIN    = 6;
14 const int RESET_PIN    = 4;
15
16 Servo flowerServo;
17 const int SERVO_STOP_DEG = 90; // Stopped
18 const int SERVO_OPEN_DEG = 135; // Rotate Right / Move Up (Opening speed)
19 const int SERVO_CLOSE_DEG = 45; // Rotate Left / Move Down (Closing speed)
20
21 const unsigned long OPEN_DURATION = 9000; // Time to fully open (ms)
22 const unsigned long CLOSE_DURATION = 9000; // Time to fully close (ms)
23
24 bool isRunning = false;
25 bool isTouching = false;
26 unsigned long touchStartTime = 0;
27 const long HOLD_TIME = 50;
28
```

Touch : High(1)
Untouch: Low(0)

Serial Monitor

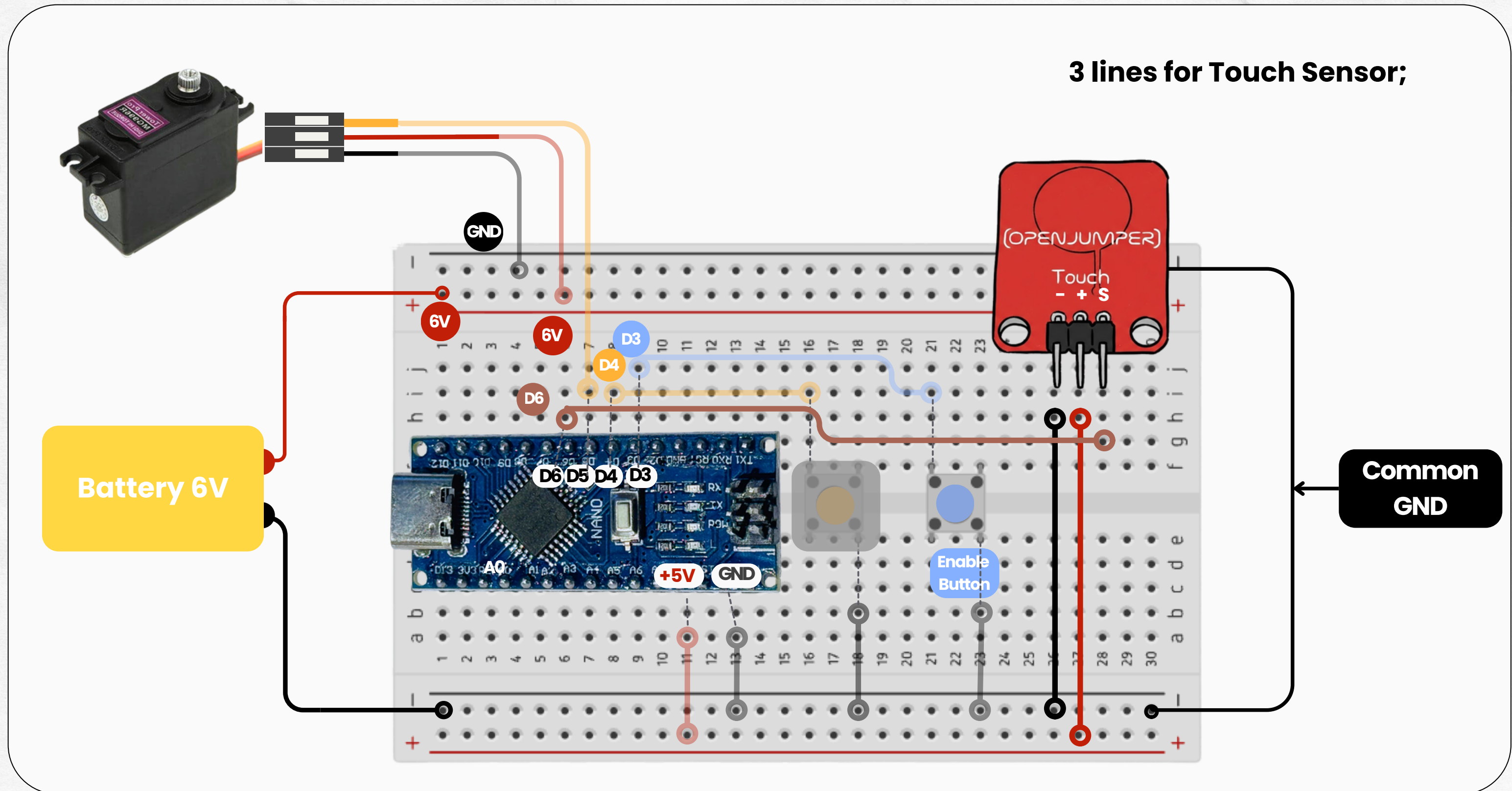


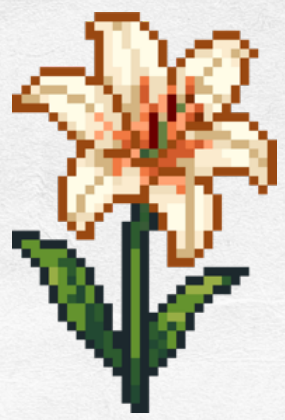
Try to use press the touch sensor and
see what happen on your Serial Monitor?



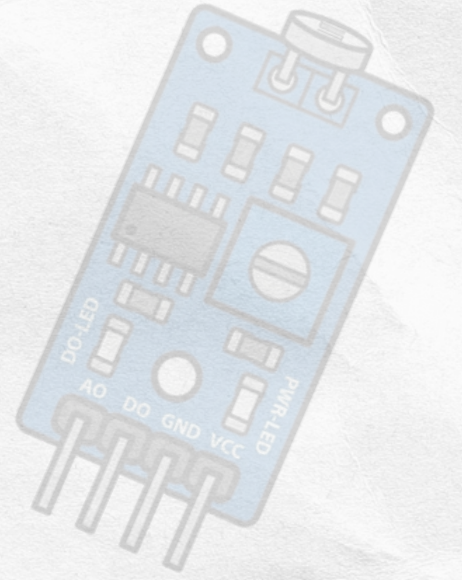
Let's Bloom Flower with Interaction!

1 Follow the diagram to connect:





Let's Bloom Flower with Interaction!



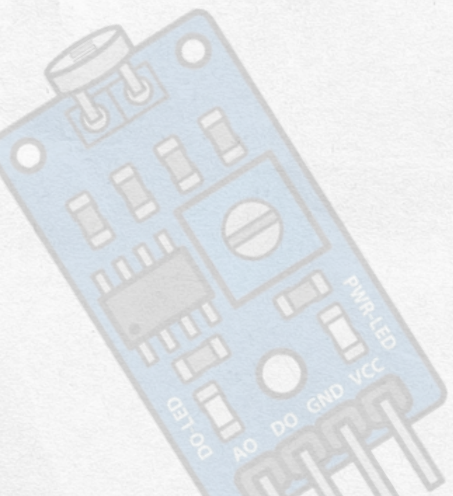
2

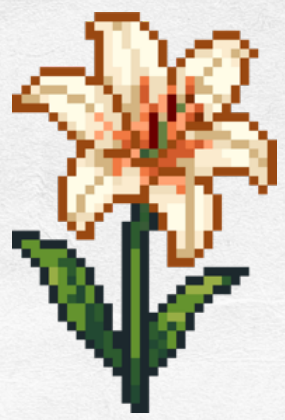
Connect **Nano** with **Laptop** by data cable;
Create a new file to paste “MF_Touch_MG996”, then Upload

 MF_Touch_MG996



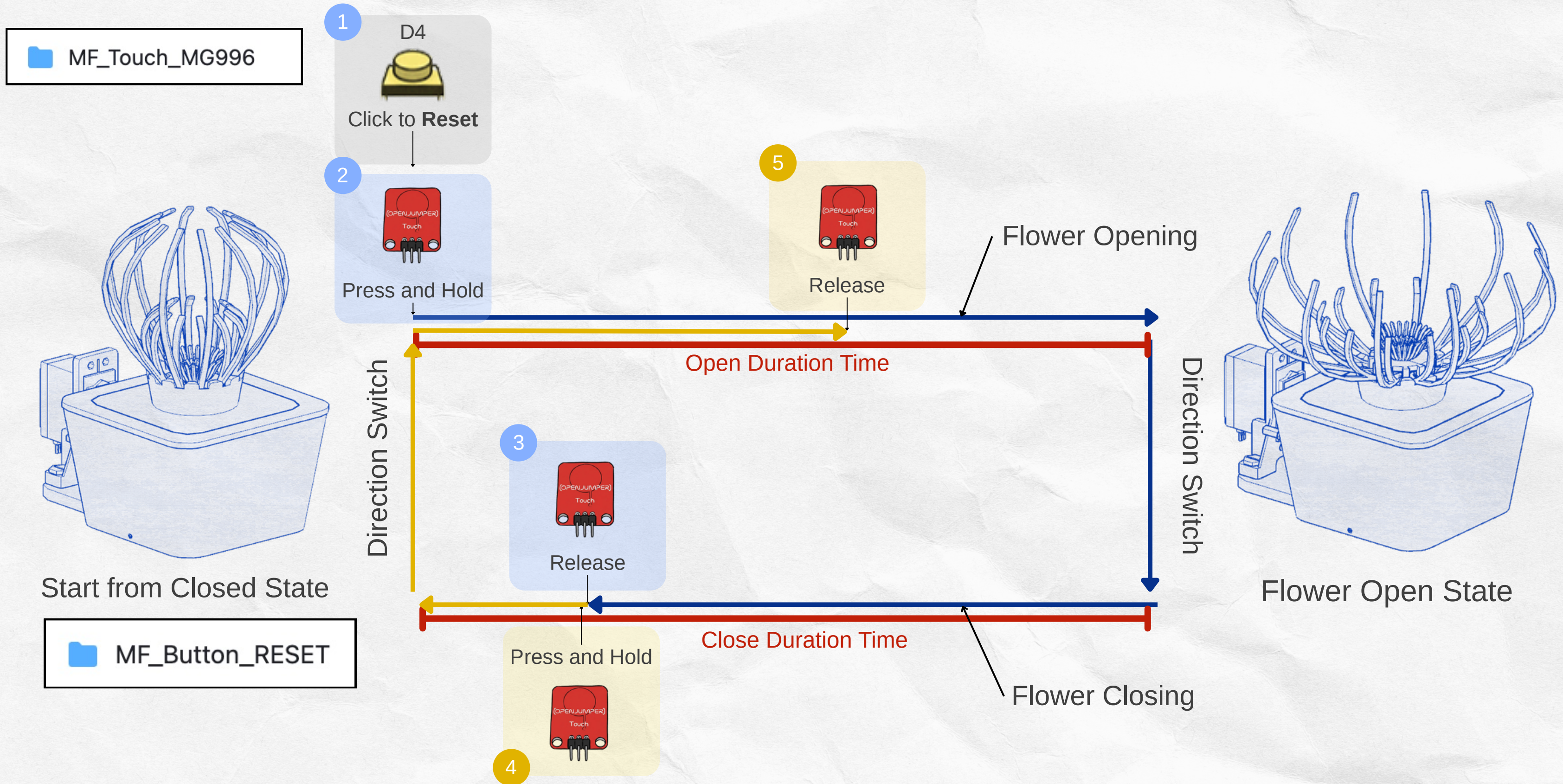
Link: <https://shorturl.at/78yy2>





Let's Bloom Flower with Interaction!

3 Touch Sensor Control the Servo — Understand Control Logic



Design a new Mechanical Flower By your imagination!



Mechanical Design: What other mechanical configurations could achieve the same movement?

Integration & Packaging: How might we design an enclosure to integrate the breadboard and mechanical components into a single unit?





Thank you for your participation!

3D Model: Max

Coding: Wickey

Helpers: Miffy Joey Emma Mo Ryan Leonie Theo